

Website Sales & Performance Analysis with SQL

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Agenda

Company Overview

Challenges and Objectives

Methodology

Traffic Source Analysis

Website Performance Analysis

Analyzing Seasonality & Business Patterns

Product Analysis

User Analysis

Get in Touch

Company Overview

Maven Fuzzy Factory is a fictional E-Commerce **retail company**, focused on **stuff animal sales**. The Company been **launched in March 2012** has been **operational for three years** with the providence of its website activities tracking the customers sessions of sales and the generated revenue.

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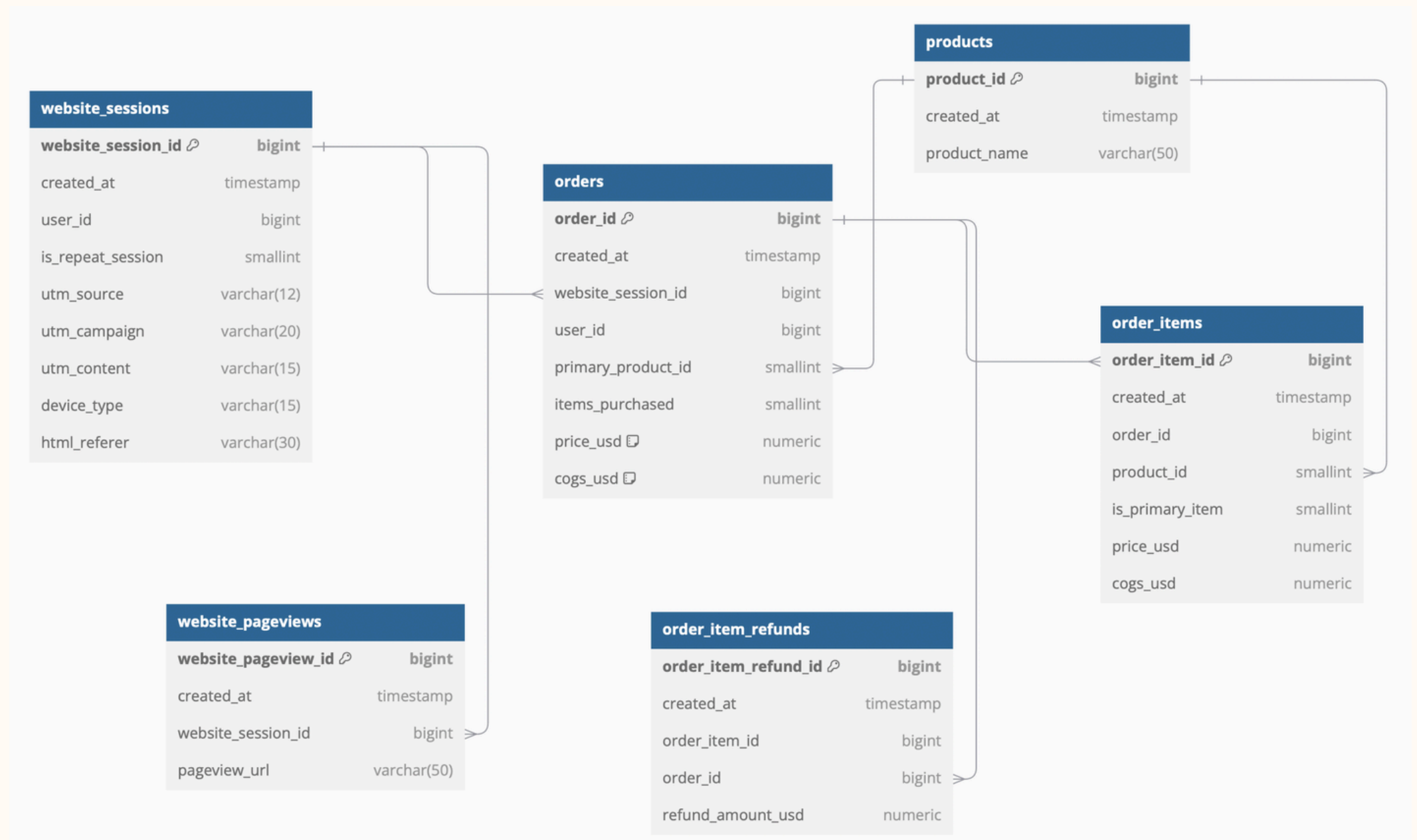


Mavenfuzzyfactory Database

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- Mavenfuzzyfactory is a relational database comprising of six tables namely, **orders**, **order_items**, **order_item_refunds**, **products**, **website_pageviews** and **website_sessions**.
- The tables **website_pageviews** and **website_sessions** comprises roughly **1 million** and **0.5 million rows**. These two tables are going to be main tables and insights would be generated by joining them with **orders**, **order_items**, **order_item_refunds** and **products** tables.

Schema!



Challenges and Objectives

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Challenges

- Extract and Analyze traffic and website performance data to craft a growth story that can enhance the company's success and development.

Objectives

- Analyse Traffic Sources
- Analyse Website Performance
- Analyse Channel Portfolio Management
- Analyse Business Pattern and Seasonality
- Product Analysis
- User Analysis

Methodology

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01

Traffic Source Analysis

- Perform overall analysis of where customers are coming from.

02

Website Performance Analysis

- Perform website analysis to understand and improve customer experience.

03

Analyzing Business Patterns & Seasonality

- Understanding key trends and business flow, to pick insights on peak seasons.

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04

Product Analysis

- Perform overall analysis of where customers are coming from.

05

User Analysis

- Understanding user behaviour is important for improving experience and business growth.

- [Finding Top Traffic Sources](#)
- [Top Traffic Sources and CVR](#)
- [Bid Optimisation](#)
- [Bid Optimization for Paid Traffic](#)
- [Results](#)

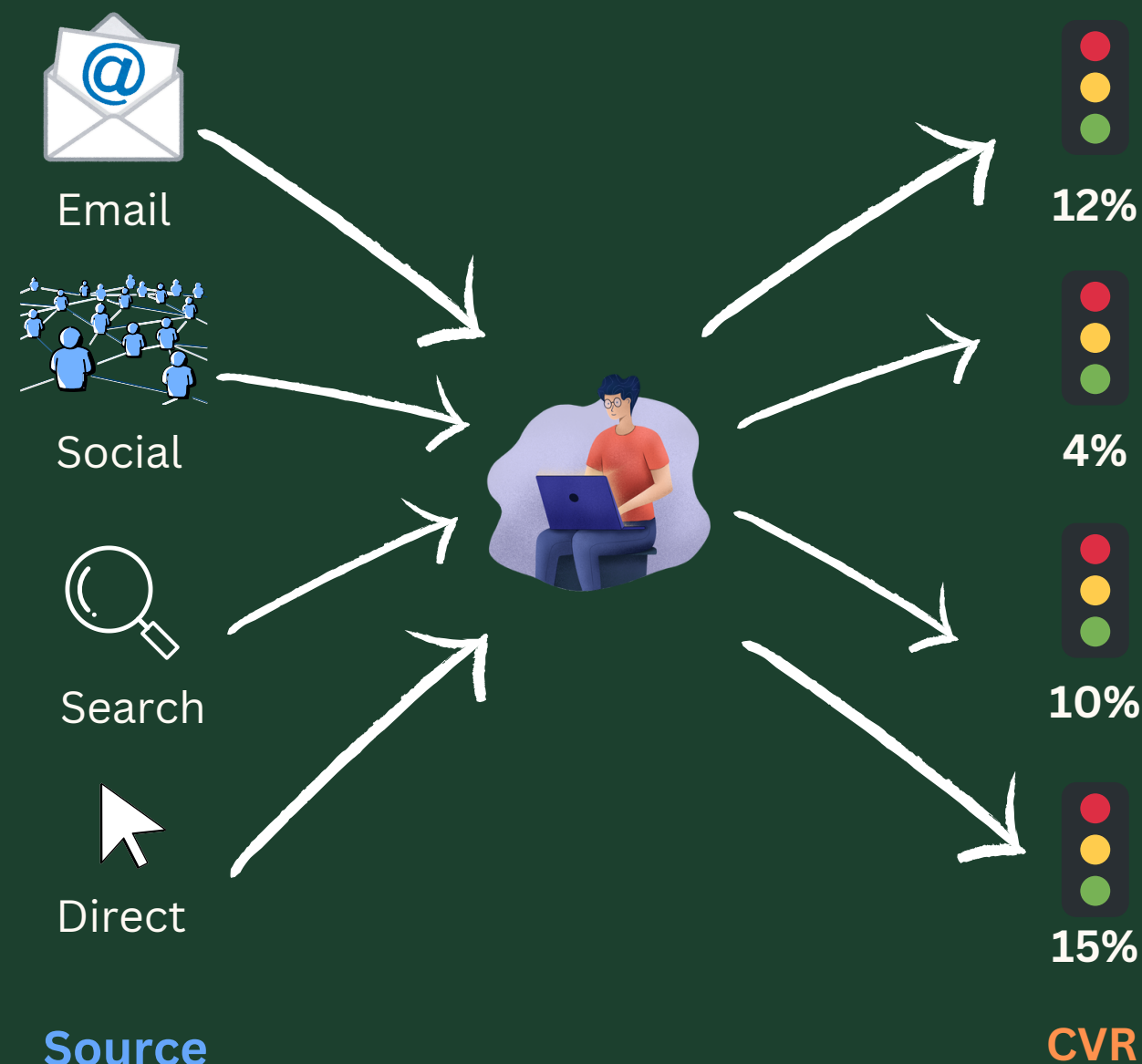


Finding Top Traffic Sources

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- Traffic source analysis is about understanding **where customers are coming from** and **which channels are driving highest quality traffic**.



USE CASES:

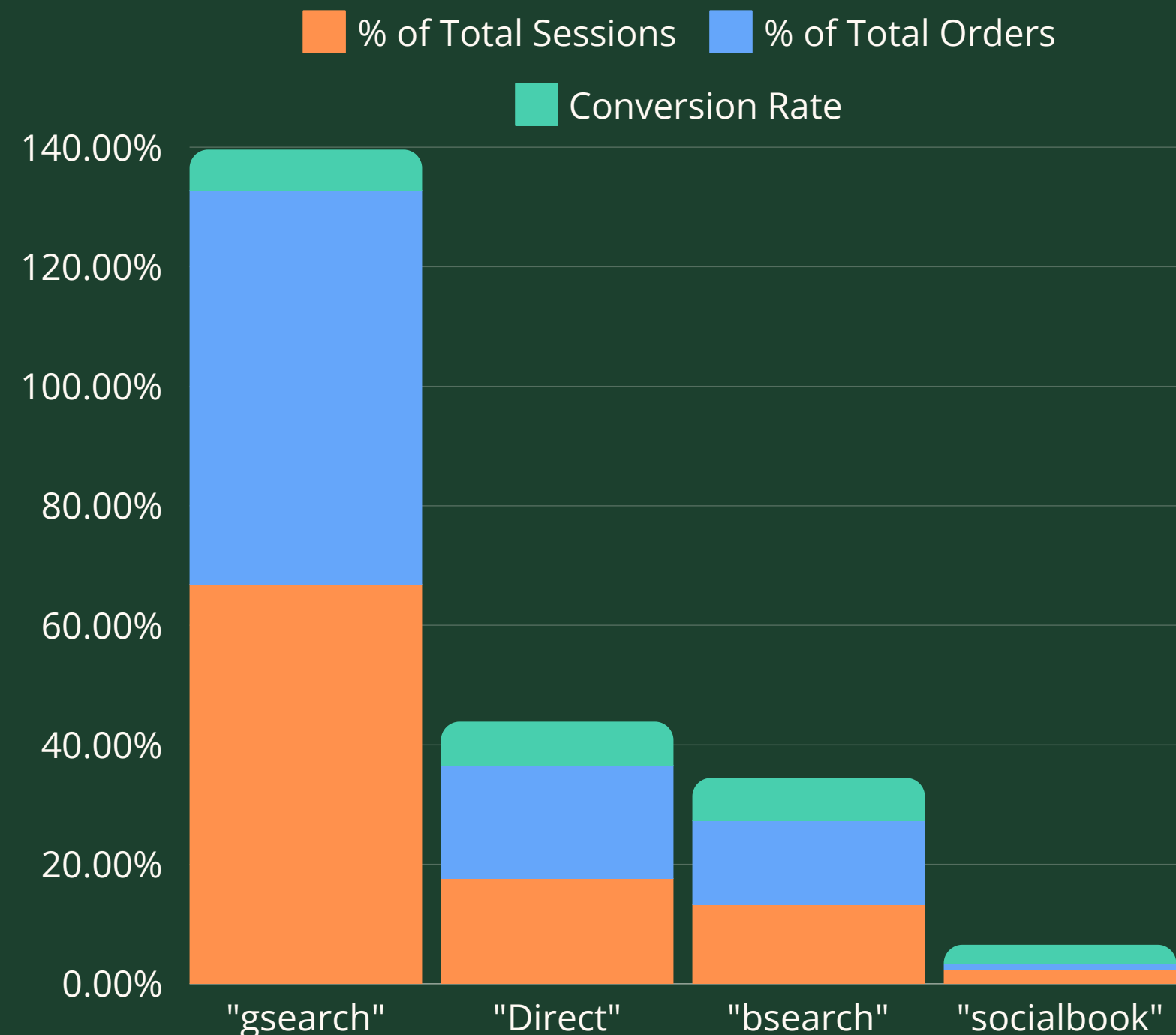
- Analysing search data and shifting budget towards engines, campaigns or key words which drive the highest conversion rates (CVR).
- Compare customer behaviour across traffic sources to improve marketing.
- Scaling high converting traffic sources.

Top Traffic Sources and CVR

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Majority of Session are coming from **Google Search**.

- Roughly 67% of session are coming from Google search, followed by Direct to website (~18%) and bsearch (~13%) and lowest via social media (~3%).
- Similar pattern is observed can be observed in total orders.
- Conversion Rates (CVR) is roughly same between all channel, i.e., ~7%
- Clearly, social media brings lowest sessions, overall orders and conversion rates (~3%)

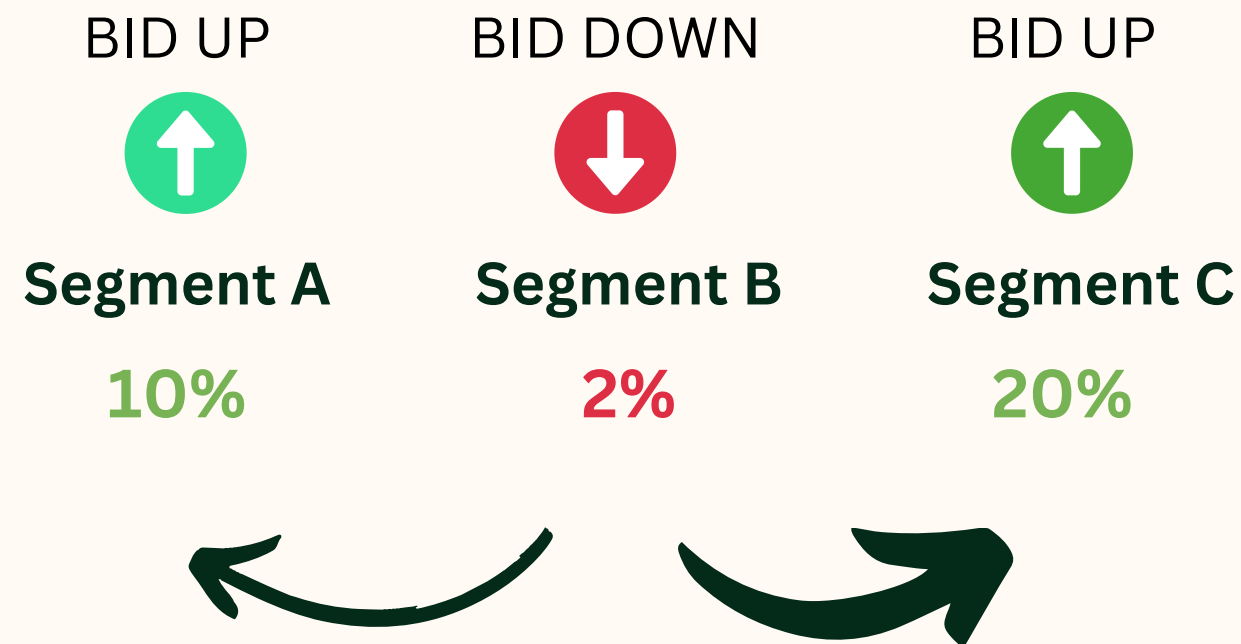


Bid Optimisation

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- Analysing for Bid Optimisation is **understanding value of various segments of paid traffic** and **optimise marketing budget accordingly**.



USE CASES:

- Using CVR and revenue per click, how much we need to spend per to acquire customers can be figured out.
- Understanding how website and products perform for various sub-segments (mobile vs desktop) to optimise the bids within these channels.
- Quantifying impact of bid changes on overall volume of customers driven to website.

BID OPTIMISATION FOR PAID TRAFFIC

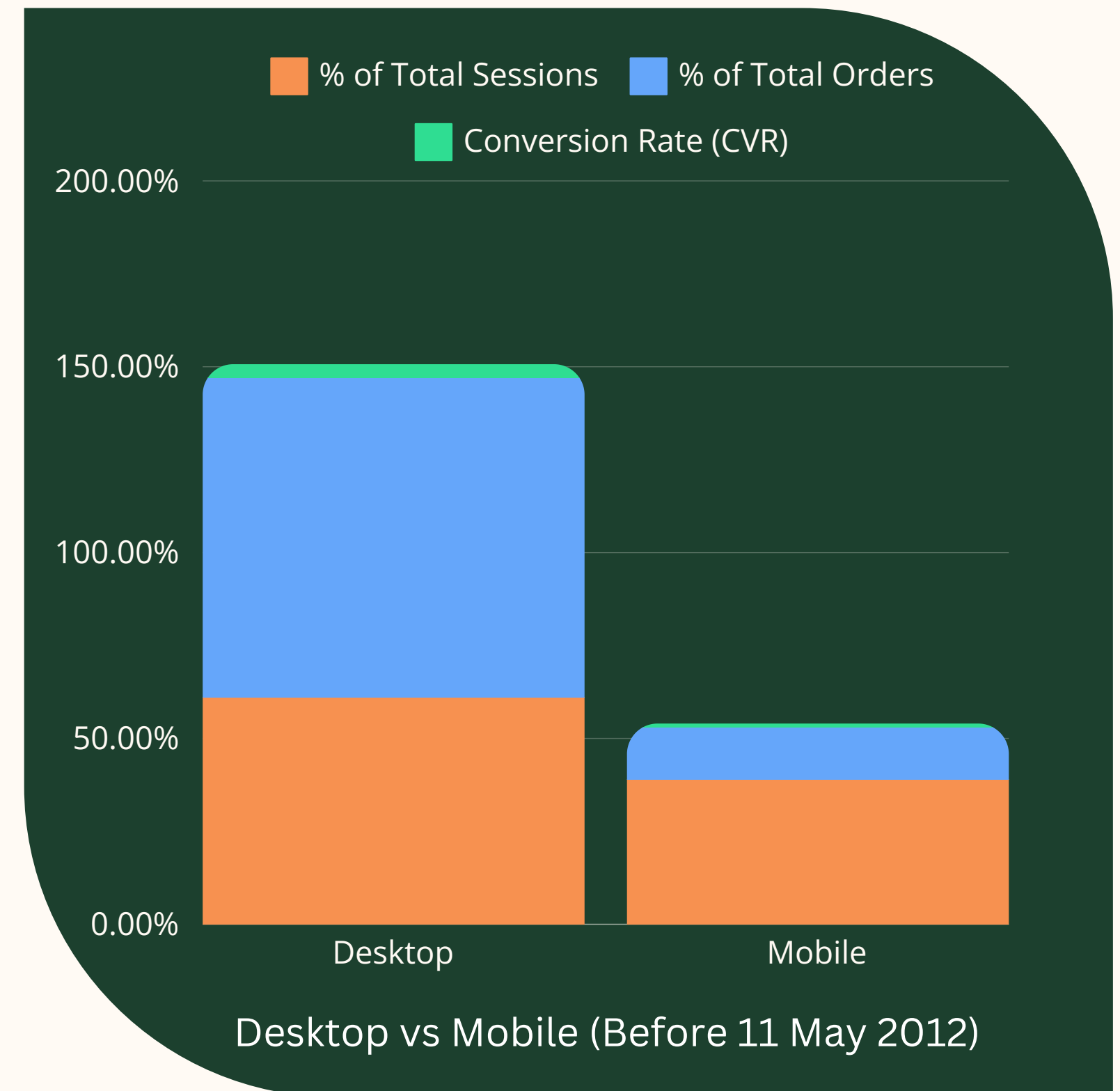
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PROBLEM: In our data, before 14 April 2012, Conversion Rate (CVR) for **utm_source = 'gsearch'** and **utm_campaign = 'nonbrand'**, CVR was 2.85% (**below 4%**).

So, company bid down for this, on 15 April 2012 and observed at significant (~40%) drop in website sessions. So, company wanted a split by device to check where CVR is higher and bid accordingly (checked on 11 May 2012).

Desktop brings ~86% of Orders and 4x CVR than Mobile.

- Desktop bring majority of sessions (~60%) and mobile (~40%).
- But desktop leads to ~86% of orders while mobile only ~14%.
- Desktop CVR is roughly 4% and mobile is 1%



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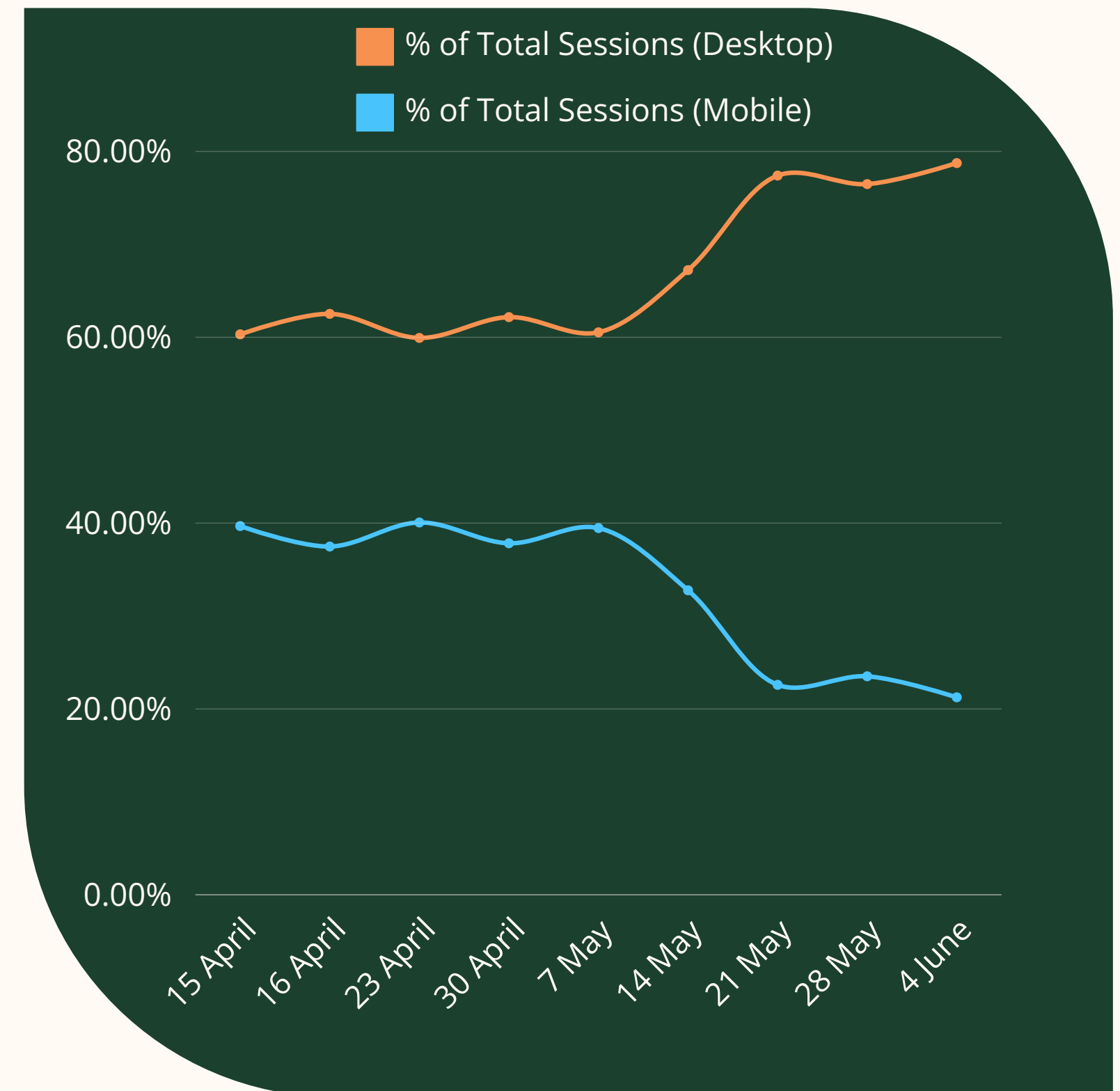
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Action: After observing that majority of traffic is brought via Desktop, company decided to bid up for desktop and bid down for mobile in gsearch nonbrand on 19 May 2012.

We need to look at performance for both segments in month.

Desktop sessions increased by ~**20%** in **four weeks**.

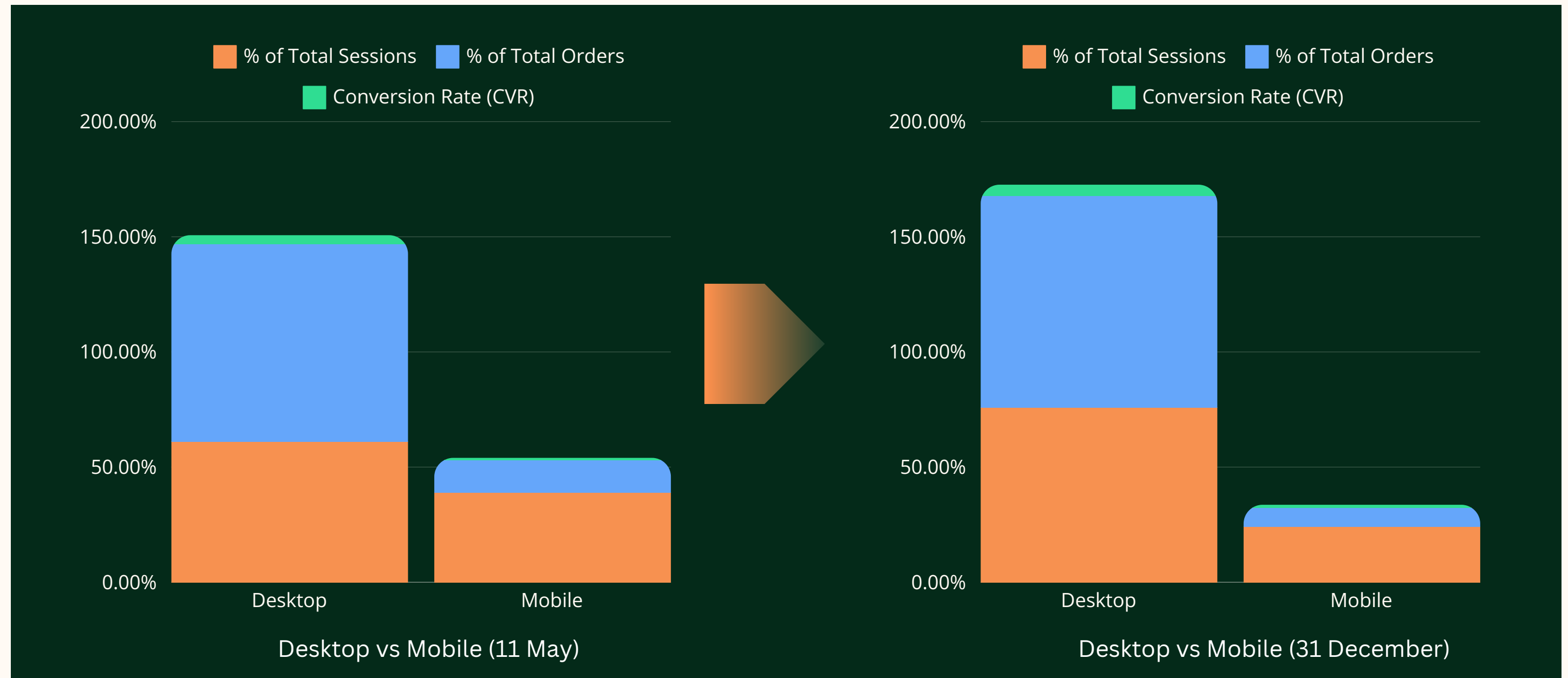
- Desktop session increased from ~60% (7 May) to ~80% (4 June)
- Mobile sessions decreased to 20% in these four weeks.



Results

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- After bidding up for Desktop in gsearch nonbrand, results were checked comparing to 11 May 2012.



- Sessions: Desktop (61%) -----> Desktop (75.87%),
- Orders: Desktop (85.88%) -----> Desktop (91.86%),
- CVR: Desktop (3.73%) -----> Desktop (4.89%),

- Mobile (38.92%) -----> Mobile (24.13%)
- Mobile (14.12%) -----> Mobile (8.14%)
- Mobile (0.96%) -----> Mobile (1.36%)

Till now, we observed that bid optimisation gave us good results for the business overall. But to improve our customer experience, we need to have a look at our website performance.

- [Analyzing Top Website Content](#)
- [Top Website Pages](#)
- [Entry Page Analysis](#)
- [Landing Page Performance Analysis](#)
- [Landing Page Bounce Rates](#)



Analyzing Top Website Content

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- Website Content Analysis helps in **understanding which pages are seen the most by our users** and **further identifying where to focus on improving business.**



Page A

550

Sessions



Page B

1800

Sessions



Page C

650

Sessions



USE CASES:

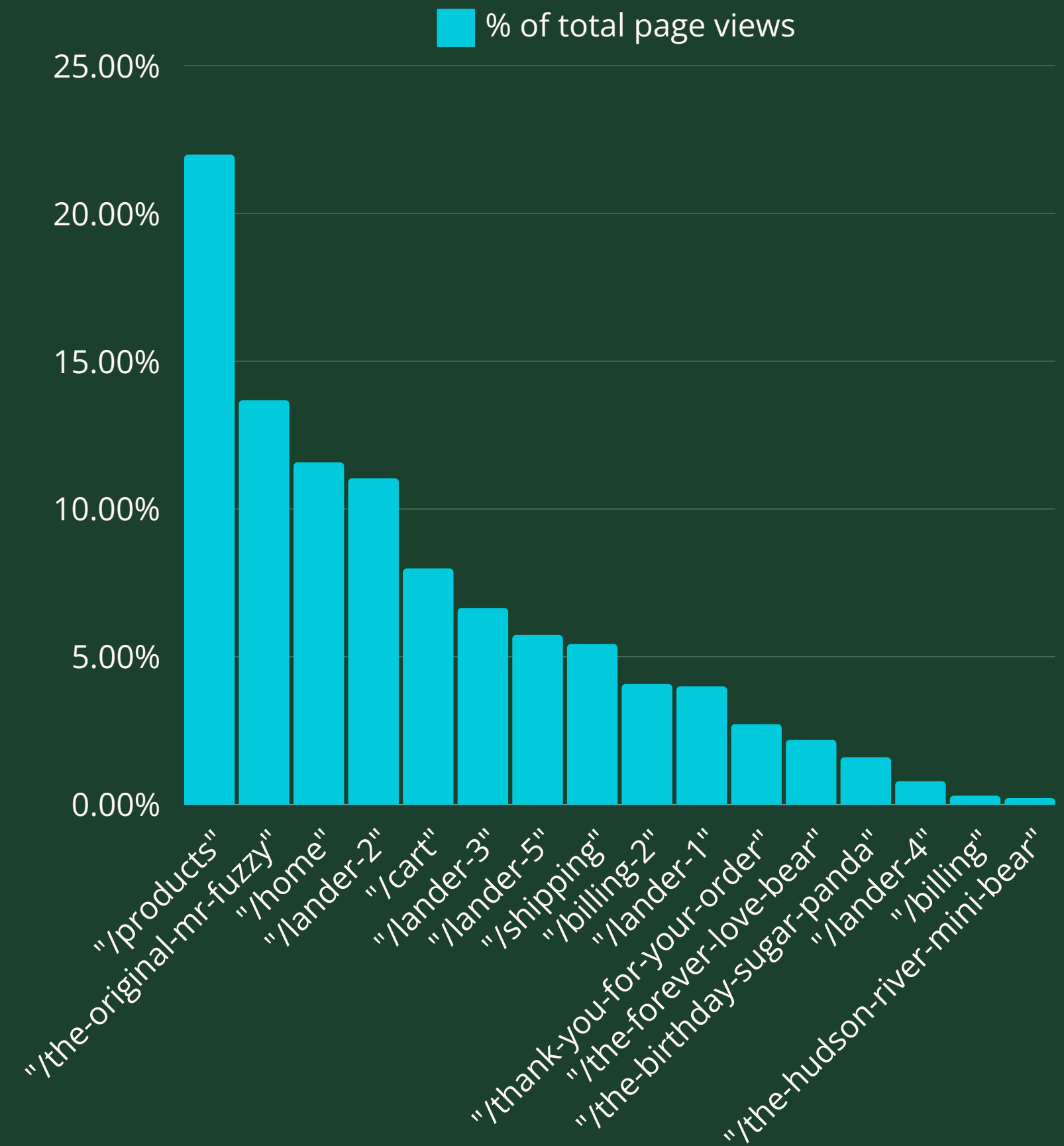
- Finding most viewed pages that customers view on website.
- Identifying most common entry pages to website-first thing the user sees.
- For most viewed pages and common entry pages, understanding how those pages perform for our business.

Finding Top Website Pages

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Products page has most of page views.

- 22% of page views are from “/products” page.
- “/the-original-mr-fuzzy” (13.68%) and “/home” are come next in page views.

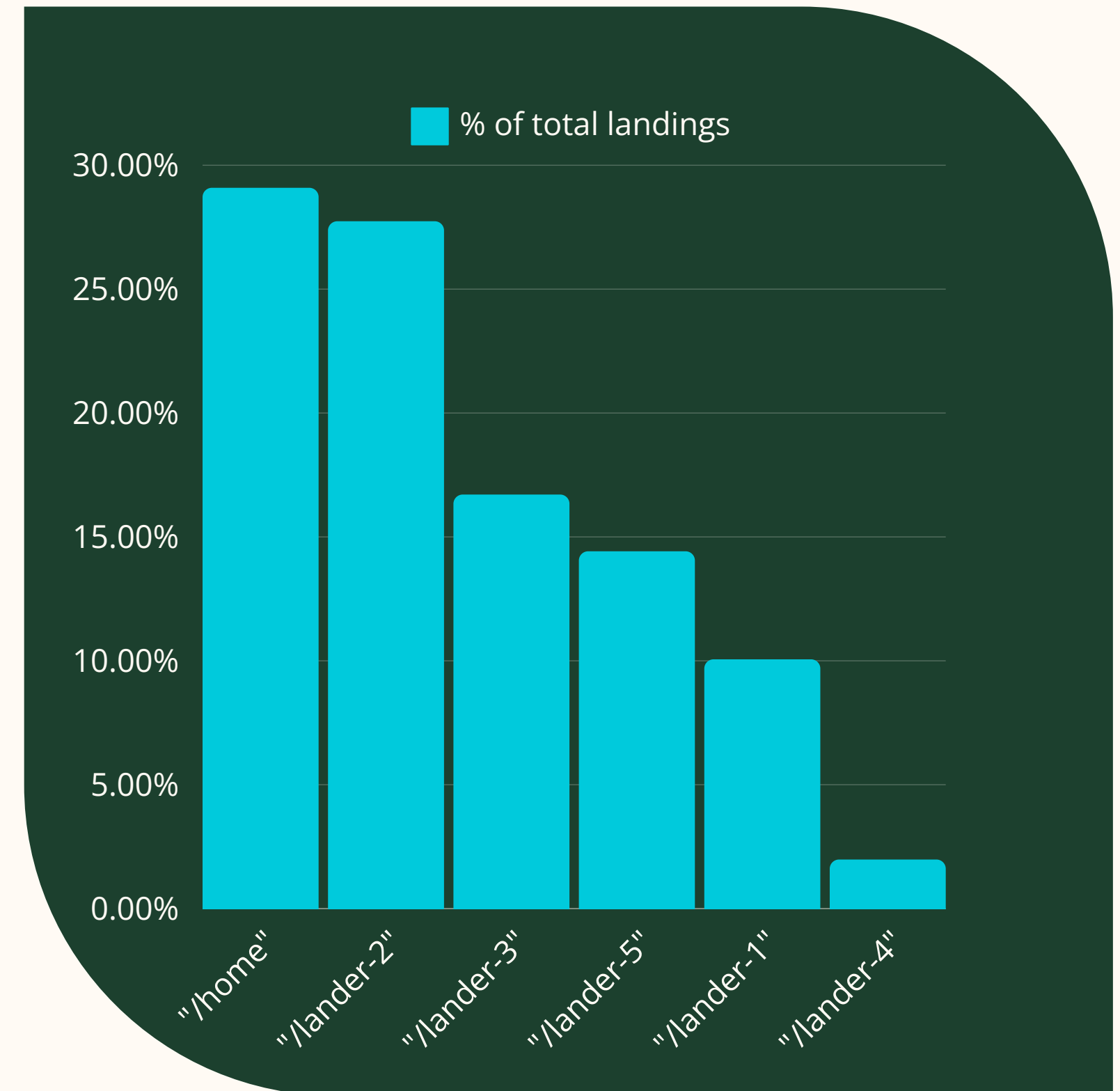


Entry Page Analysis

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Majority of users land on “/home” page

- Majority of users land on home page.
- It would be explained later that landing pages have actually been used to perform A/B Testing to experiment with new home pages.
- It is best advisable to customise home page as it is the one seen by users first time.

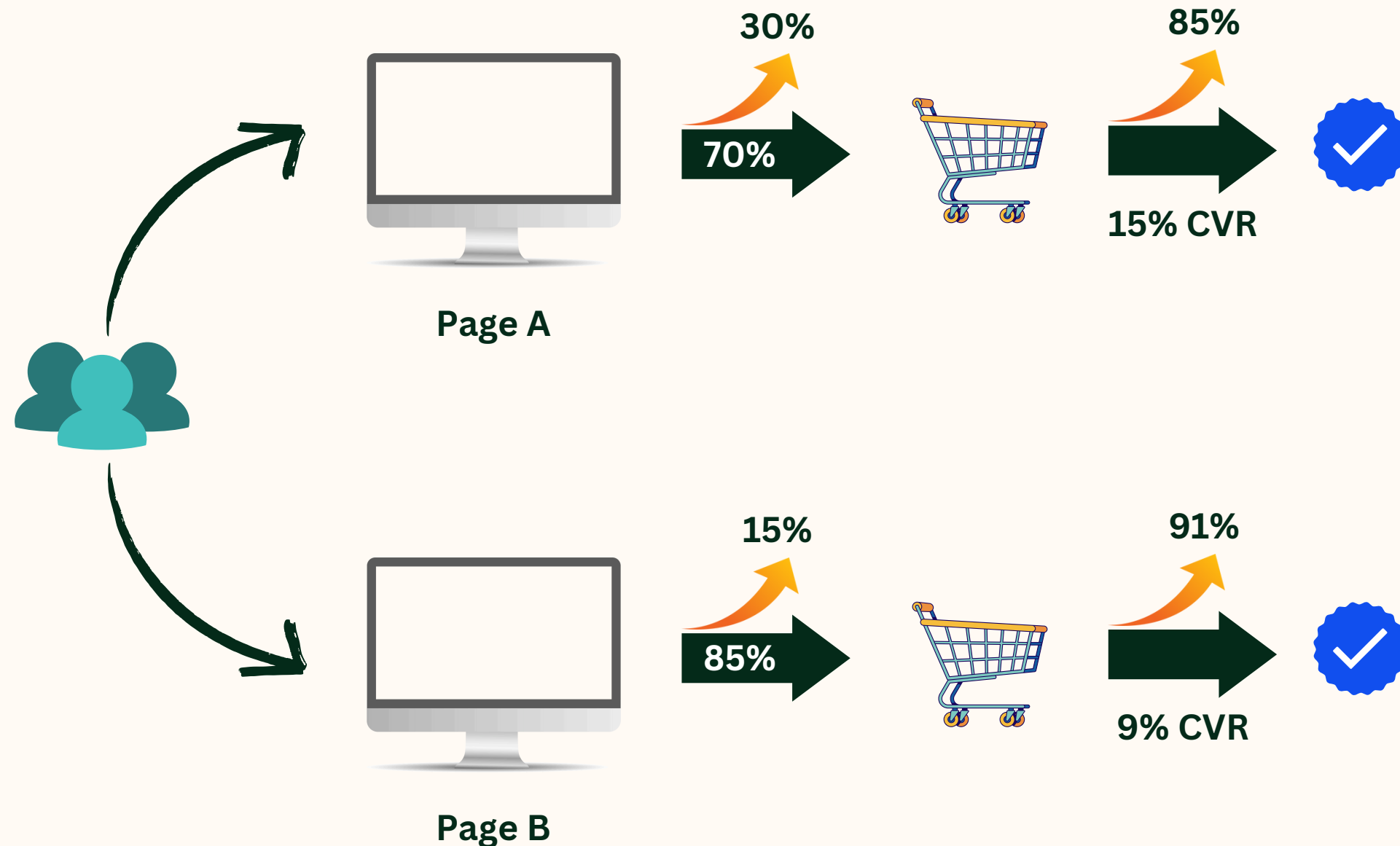


Landing Page Performance Analysis

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- Landing page analysis and testing is about understanding the **performance of key landing pages** and **then testing to improve our results**.



USE CASES:

- Identifying top opportunities for landing pages - high volume pages with higher than expected bounce rates or lower CVR.
- Setting A/B experiments on live traffic to see if we can improve our bounce rate and CVR
- Analyzing test results and making recommendations on which version of landing page to choose for going forward.

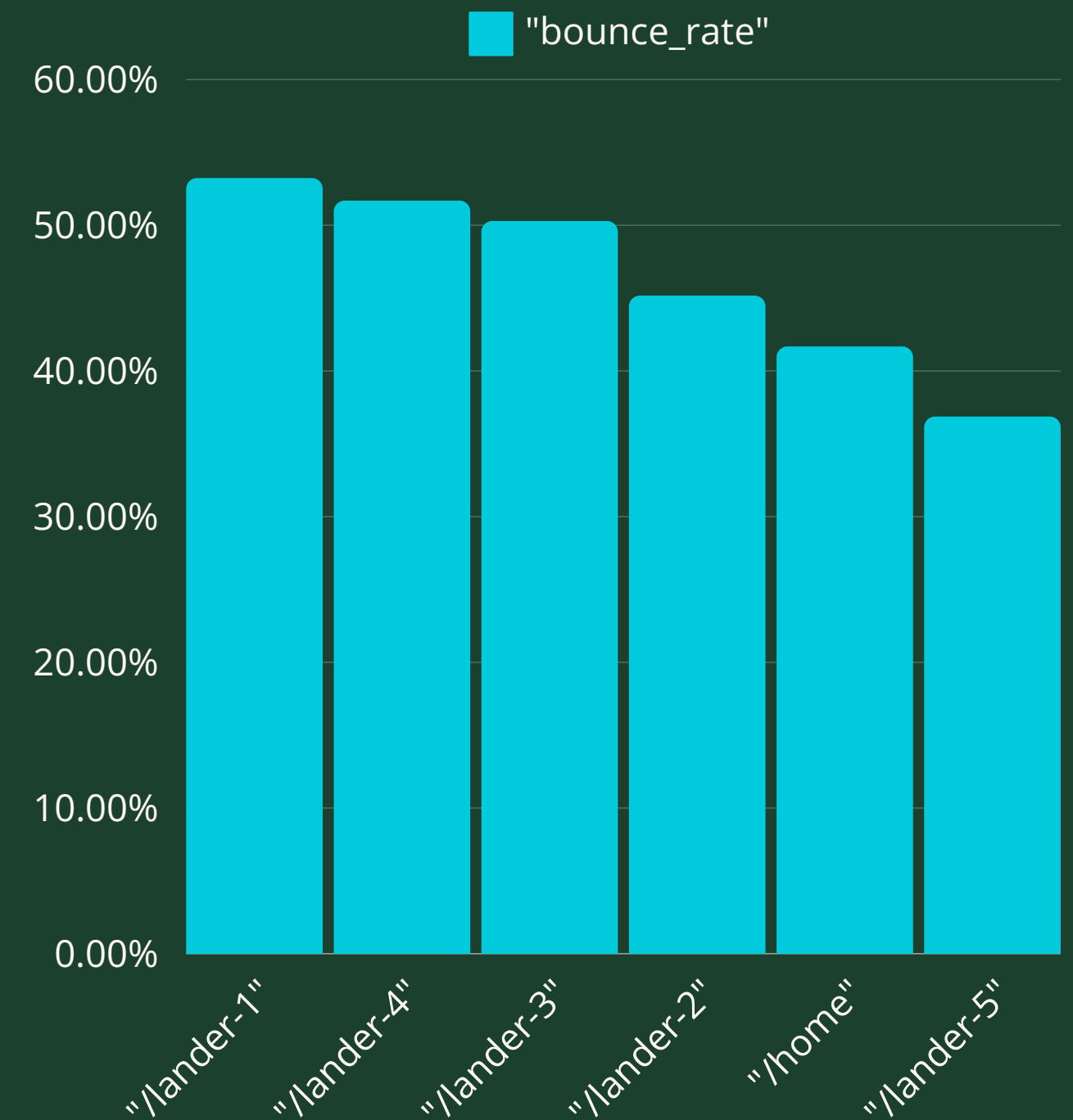
Landing Page Bounce Rates

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Only 47/100 would be able to land on **/lander-1** page on an average.

- /lander-1 has **53% bounce rate** meaning majority of customers would not be able to land on this page.

What Exactly are landing pages in our case?



50/50 Landing page Test

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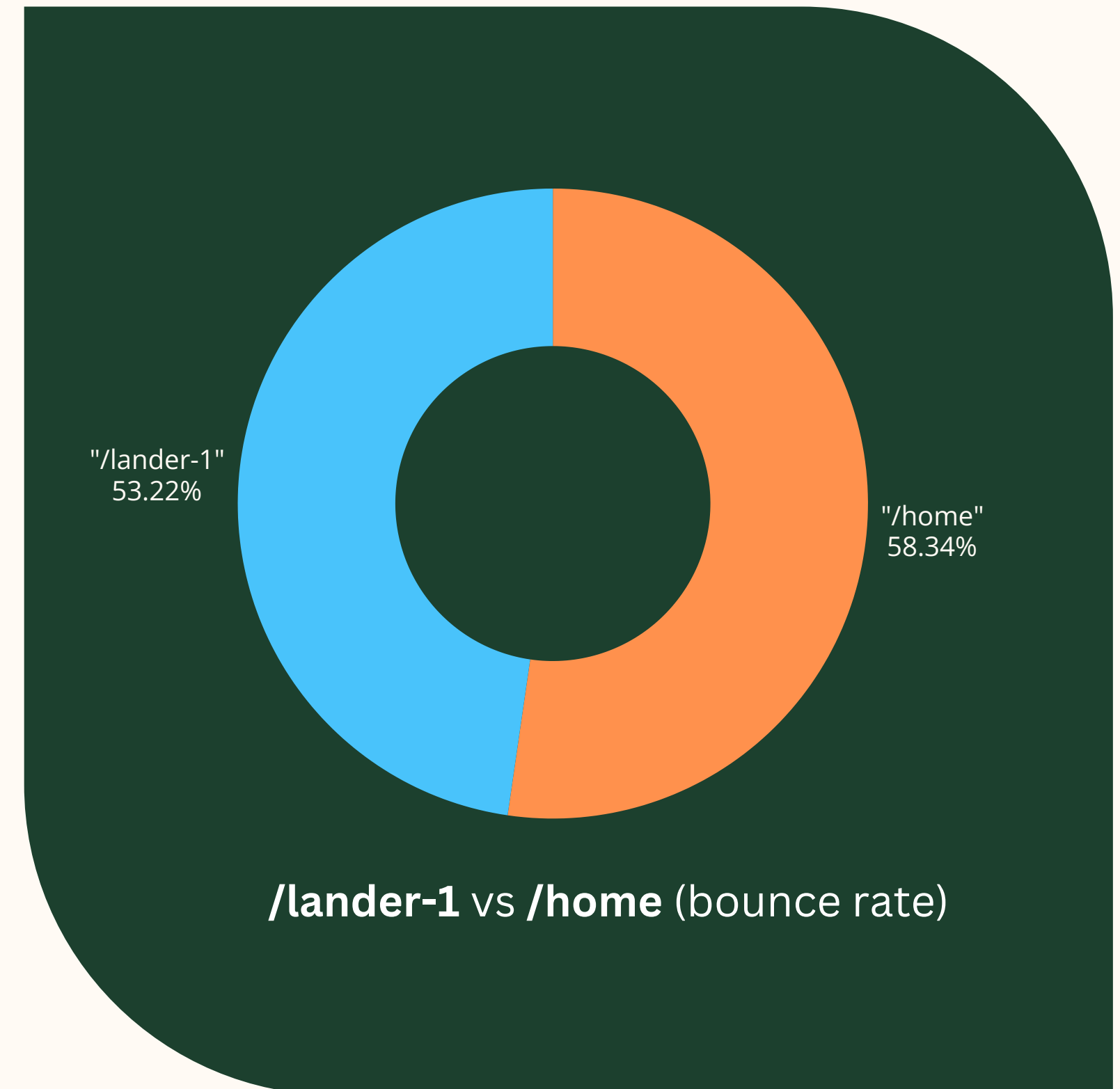
Company decided to run a **50/50 test of /lander-1 against /home for gsearch nonbrand traffic**. We are required to **evaluate bounce rates** for these two pages.

Also, we need to look at the time frame where /lander-1 was getting traffic, till July 28, 2012.

/lander-1 had lower bounce rate than /home page

- Not only /lander-1 had lower bounce rate than /home but also, it had higher number of sessions comparing to /home.
- /lander-1 = 2315 sessions
- /home = 2261 sessions.

WHAT NEXT??



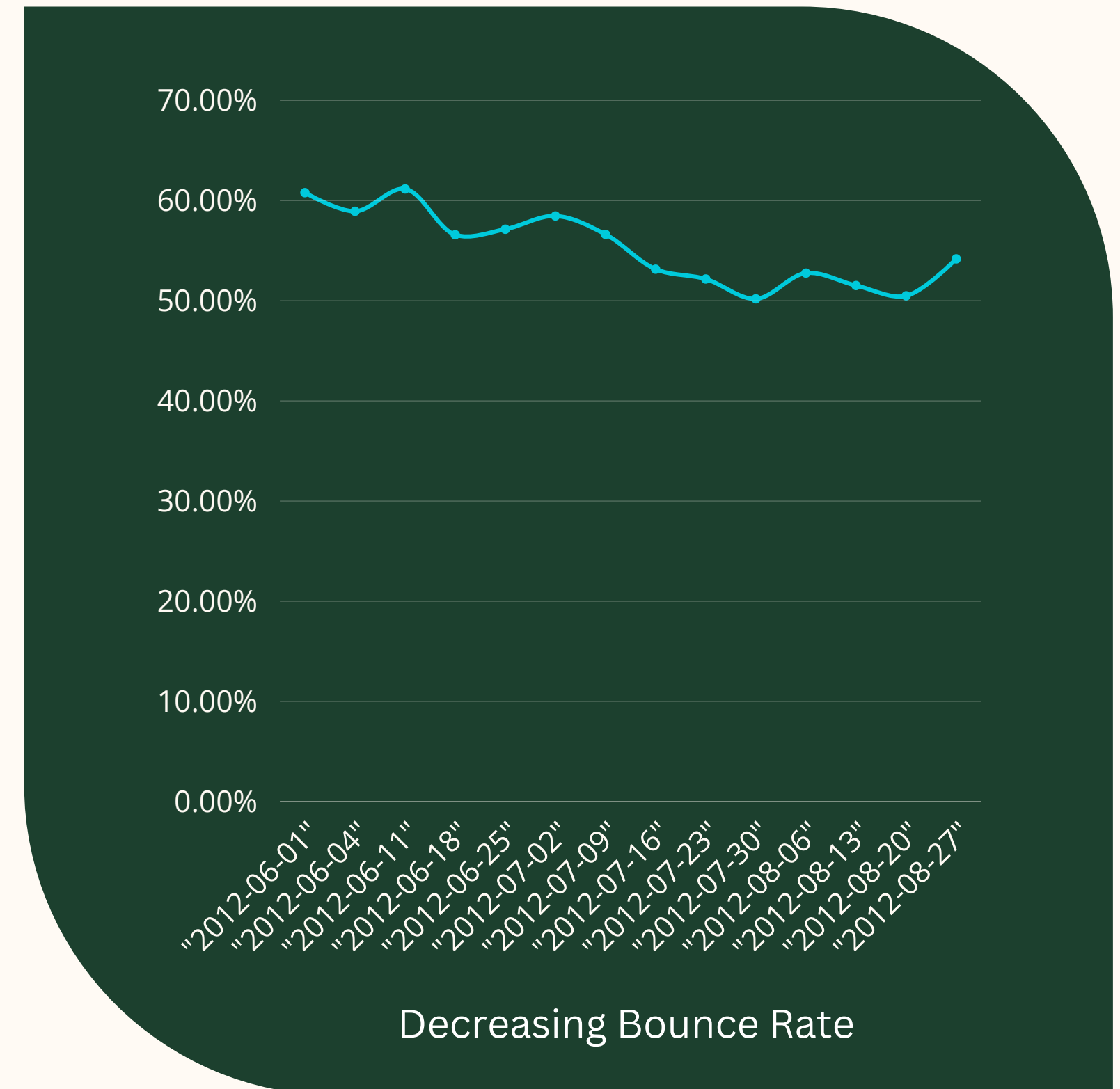
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Clearly, /lander-1 gave great results. So, all paid gsearch nonbrand traffic was routed to /lander-1 instead of /home and progress was checked for the coming weeks starting from 1 June 2012.

Bounce rate **decreased by 10%** after traffic getting **routed to /lander-1**

- Starting at ~61% (June 1, 2012), bounce rate decreased to ~51% by 20 August 2012 (in roughly 2 months).



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Analyzing Seasonality & Business Patterns

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- [Introduction](#)
- [Seasonality by Year \(Session\)](#)
- [Seasonality by Year \(Order\)](#)
- [Live Chat Support?](#)
- [Average Session by Hour](#)



Analyzing Seasonality & Business Trends

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- Analyzing business patterns is about **generating insights to help us maximise efficiency and anticipate future trends.**



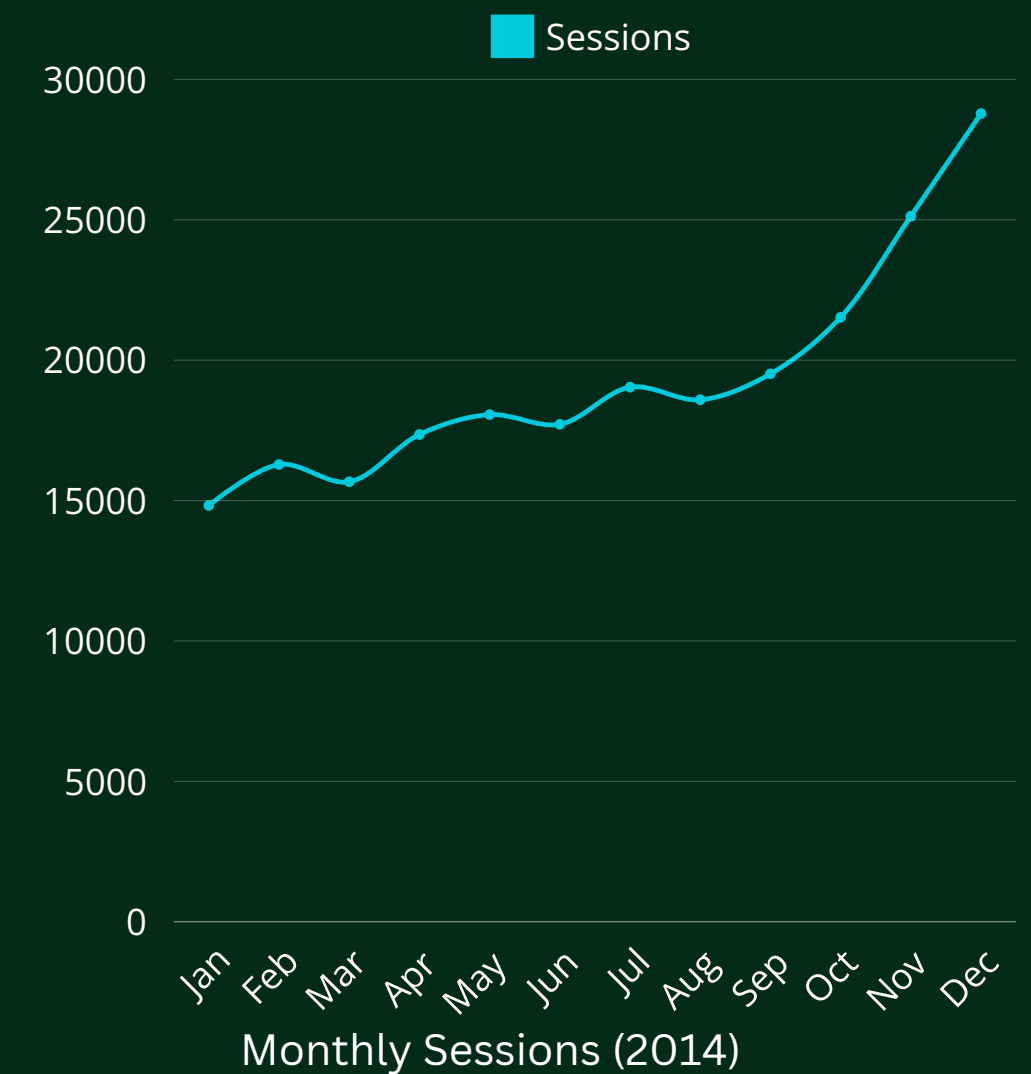
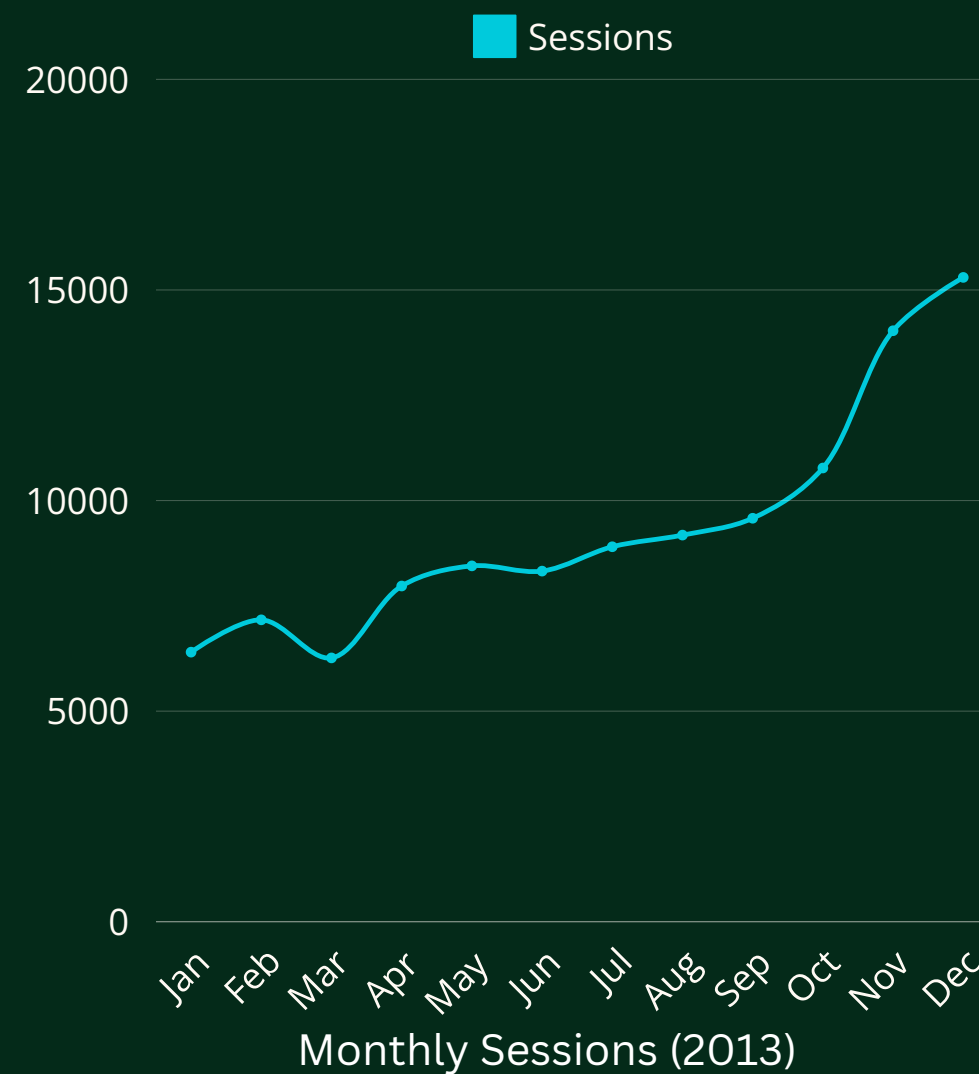
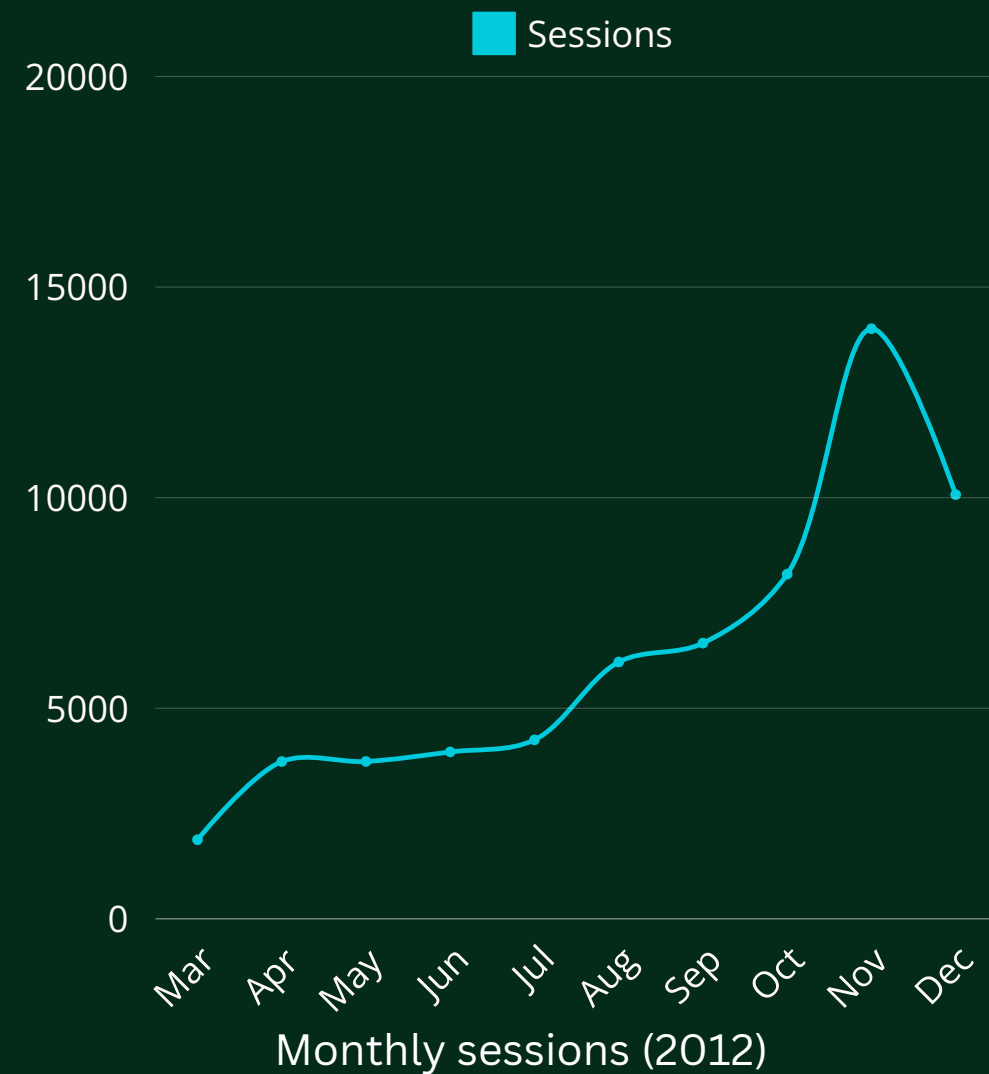
USE CASES:

- Day-parting analysis to understand how much support staff is needed at different times of day.
- Analyzing seasonality to better prepare for upcoming spikes or slowdowns in demand.

Seasonality by Year (Session)

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Total website sessions have doubled in 2 years.

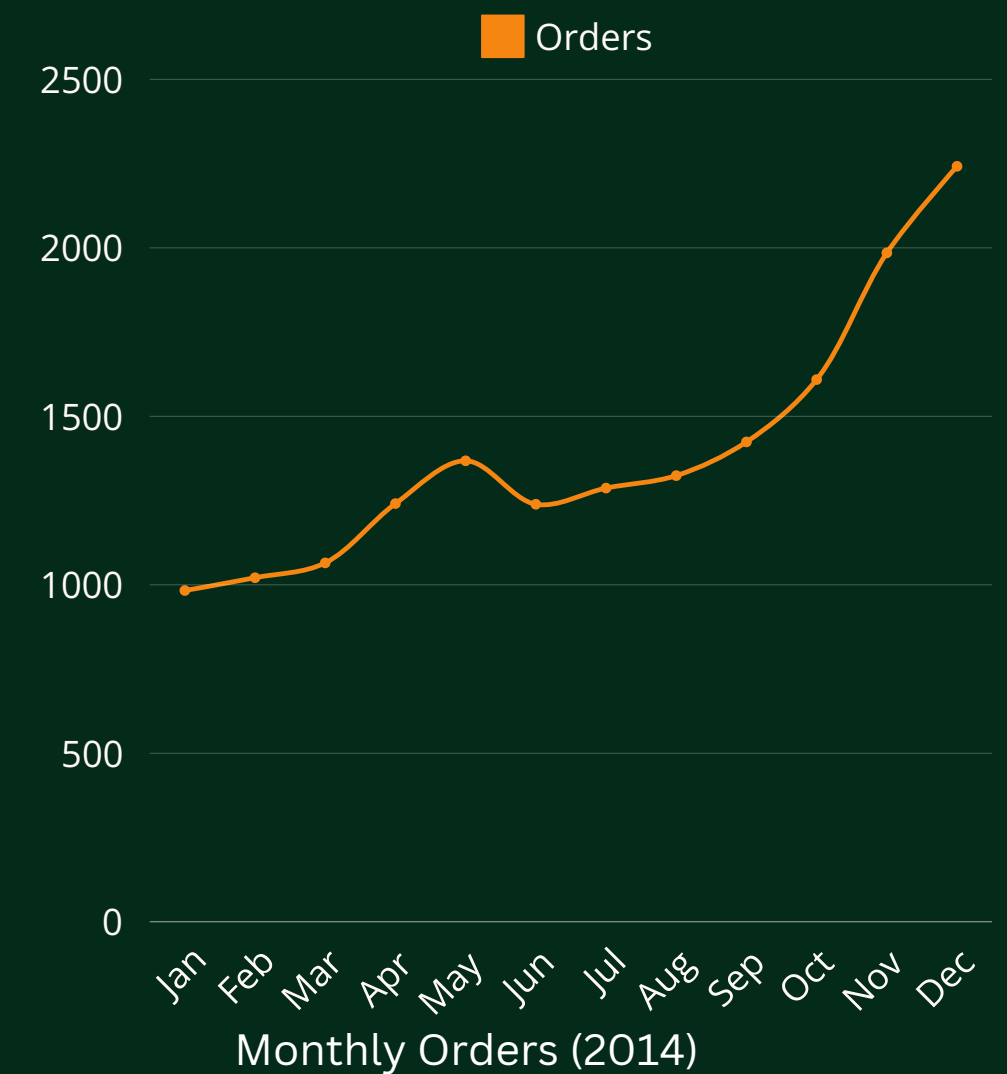
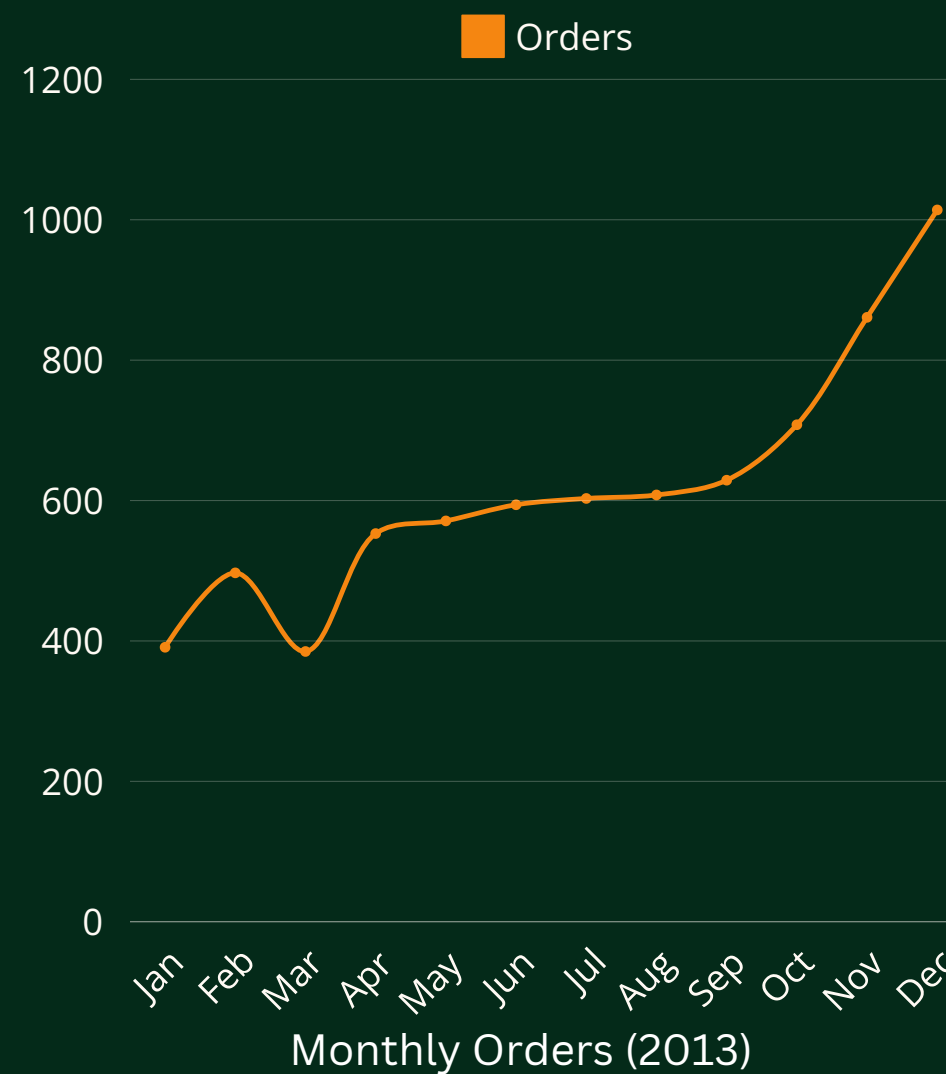
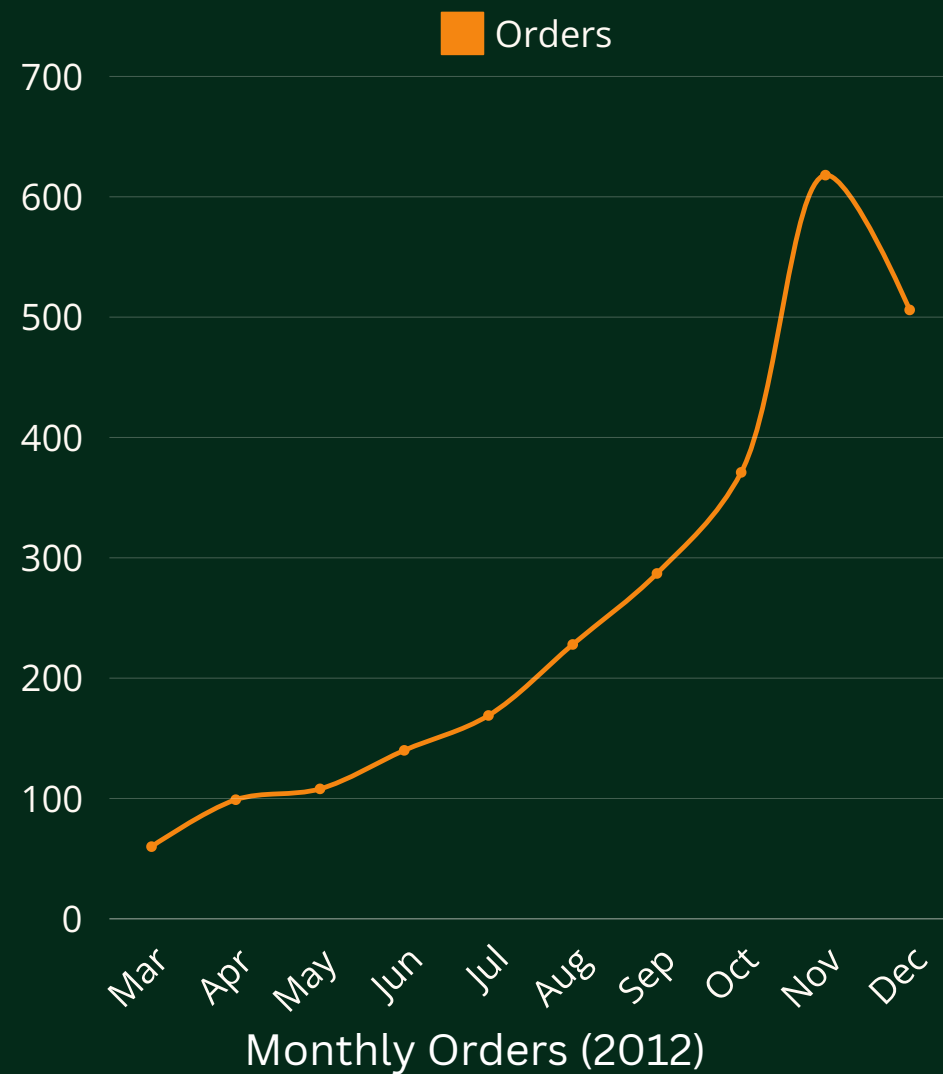


- For 2012, sessions increases as months progress peaking in November (14K sessions)
- 2013, sessions increase over months peaking in December (15K sessions)
- 2014, sessions increase over months peaking in December (28K sessions), twice the initial year.
- Overall website sessions have doubled in the span of 2 years, which is a positive indicator to business growth.

Seasonality by Year (Order)

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Total orders have tripled in 2 years. Sessions and orders have high correlation.



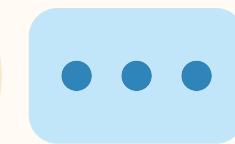
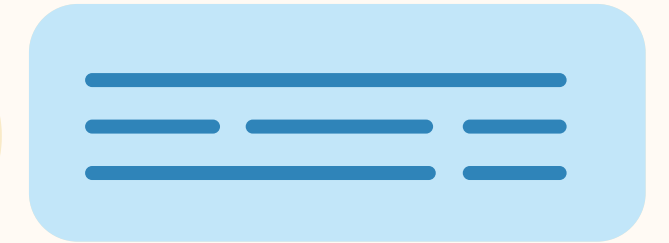
- For 2012, number of orders increase as months progress peaking in November (618 orders)
- 2013, orders increase over months peaking in December (1054 orders)
- 2014, orders increase over months peaking in December (2242 orders), thrice the initial year.
- Overall total orders have tripled in the span of 2 years, which is a positive indicator to business growth.

Live Chat Support, a good idea?

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It was observed that **fourth quarter of the year (Sep - Dec)** has **highest website session and order volume**.

So, company is planning to **add Live Chat Support system** to improve customer experience. But, **how to allocate support staff to maximise efficiency?**

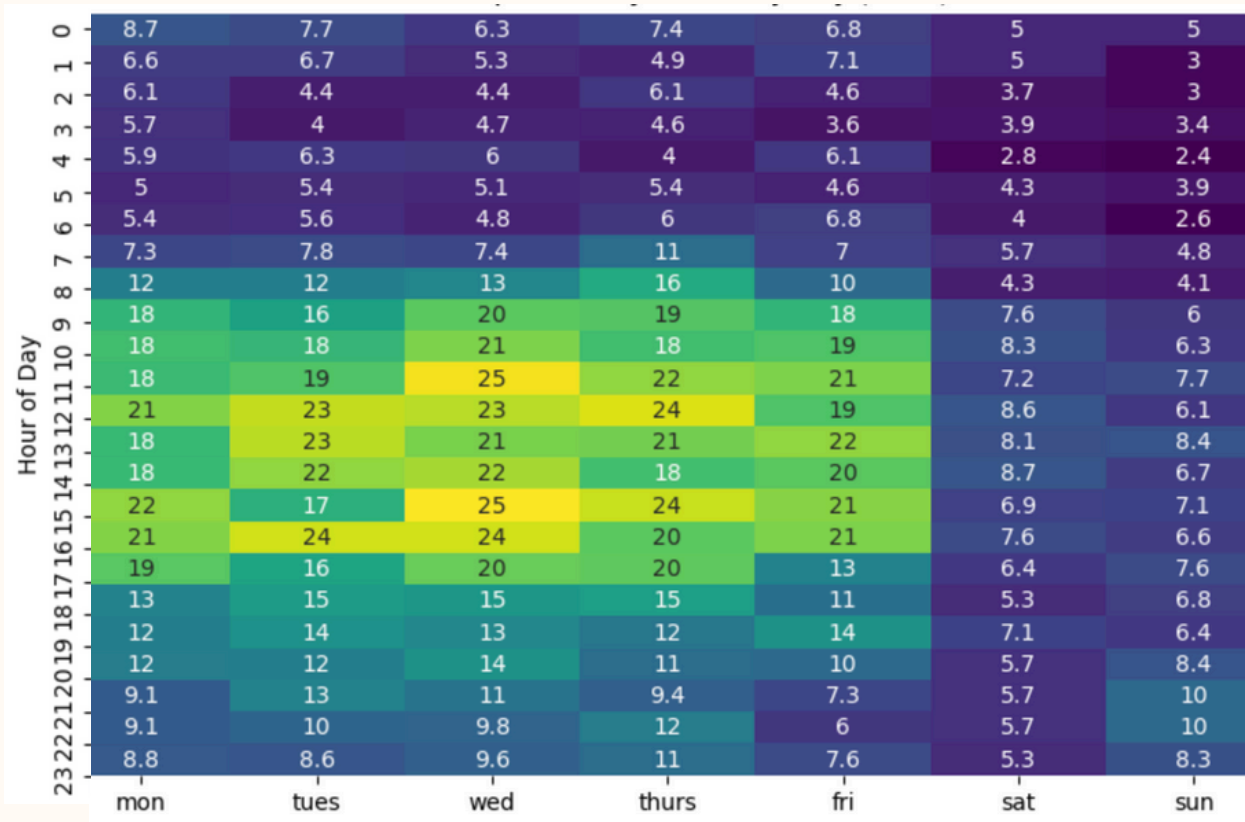


Solution: Let us map **average number of sessions by hour of day** throughout the **entire week on yearly basis**.

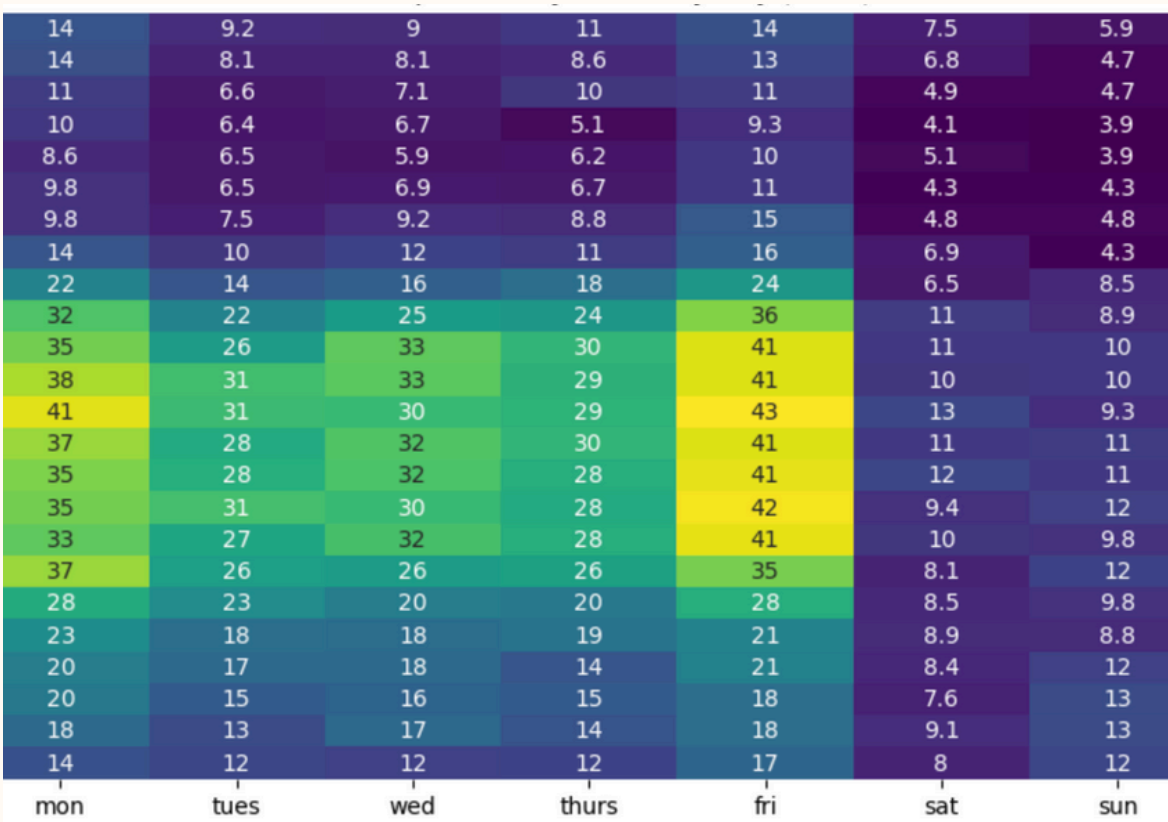
Average Sessions by Hour over Week

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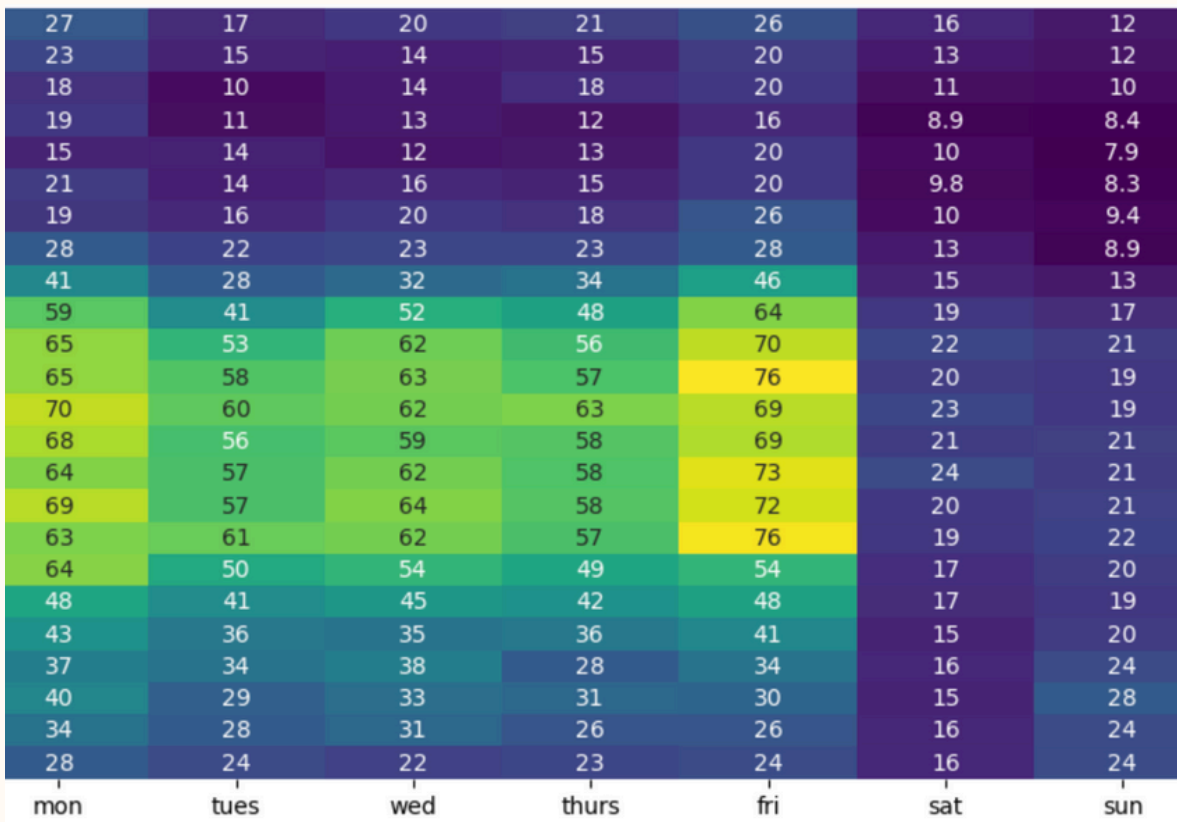
8 AM to 5 PM, Monday to Friday is ideal time for **higher support staff**.



Heatmap (2012)



Heatmap (2013)



Heatmap (2014)

- In all three years, higher number of sessions can be observed between 8 AM to 5 PM across weekdays (bright areas in heatmap).
- As average session ranges between 60-80 (in latest year, 2014) and one support staff member can handle ~10/hr, thus 6-8 number of support staff is advisable for live chat support to address customer queries.

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Product Analysis

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- [Introduction](#)
- [Product Business over years](#)
- [Revenue, Margin & AOV](#)
- [Product Level Conversion Funnel](#)
- [Monthly Sales & Revenue](#)
- [Product Level Website Analysis](#)
- [Cross Selling Analysis](#)
- [Product Refund Analysis](#)

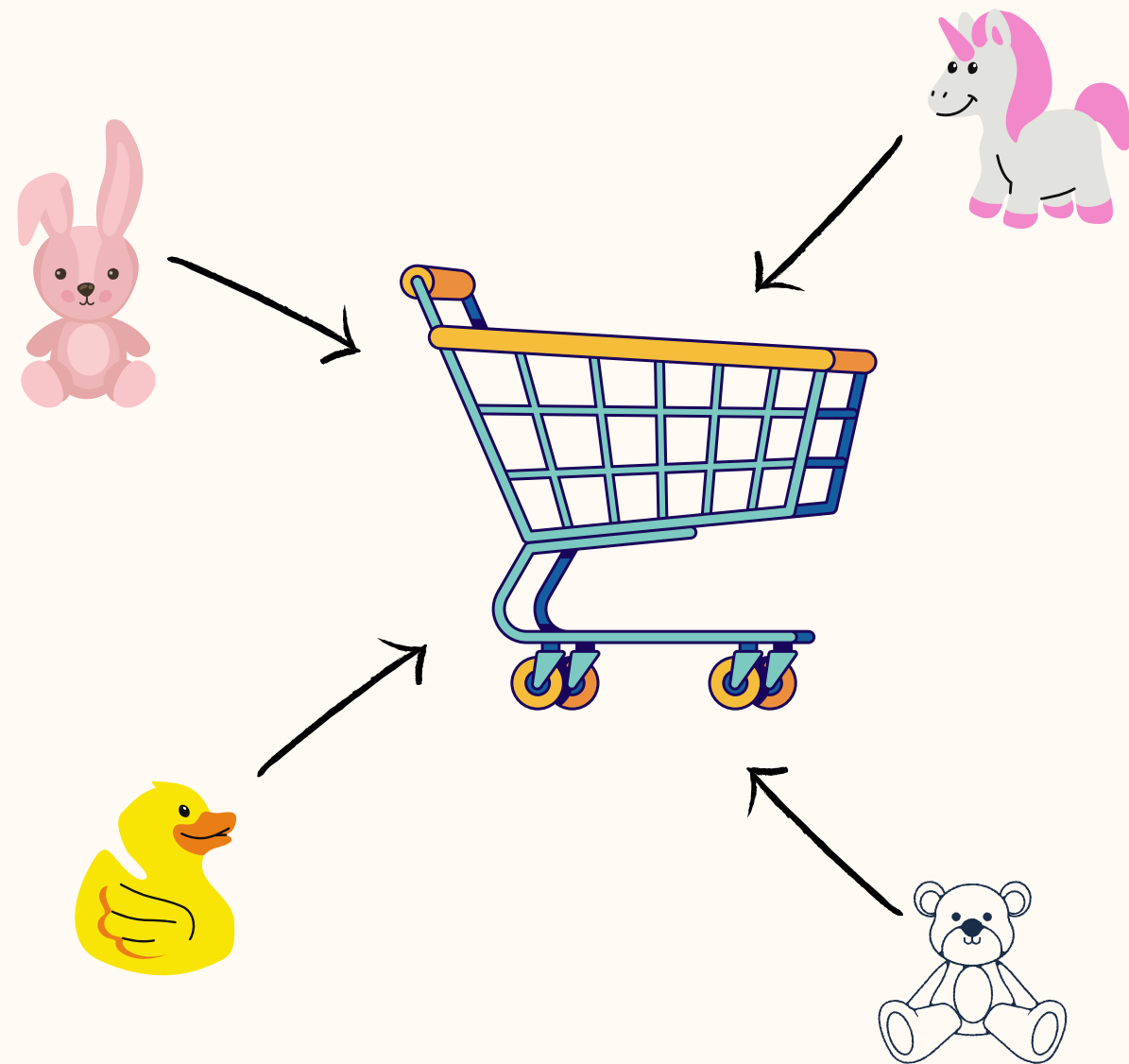


Product Analysis

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Analyzing product sales helps in **understanding how each product contributes to business** and **how product launches impact overall portfolio**.

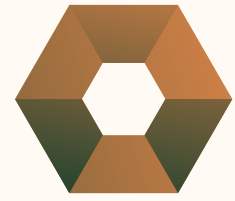


USE CASES:

- Analyzing sales and revenue by product.
- Quantifying impact of adding new product on revenue.
- Monitoring product sales trends to understand overall health of our business.

Business Terms

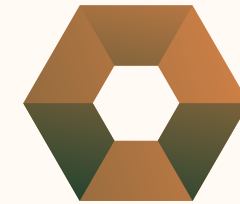
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Orders

Number of Orders placed by customers.

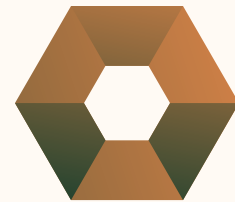
COUNT(order_id)



Margin

Revenue less cost of good sold.

SUM(price_usd - cogs_usd)



Revenue

Money coming from the business.

SUM(price_usd)



AOV

Average revenue per order.

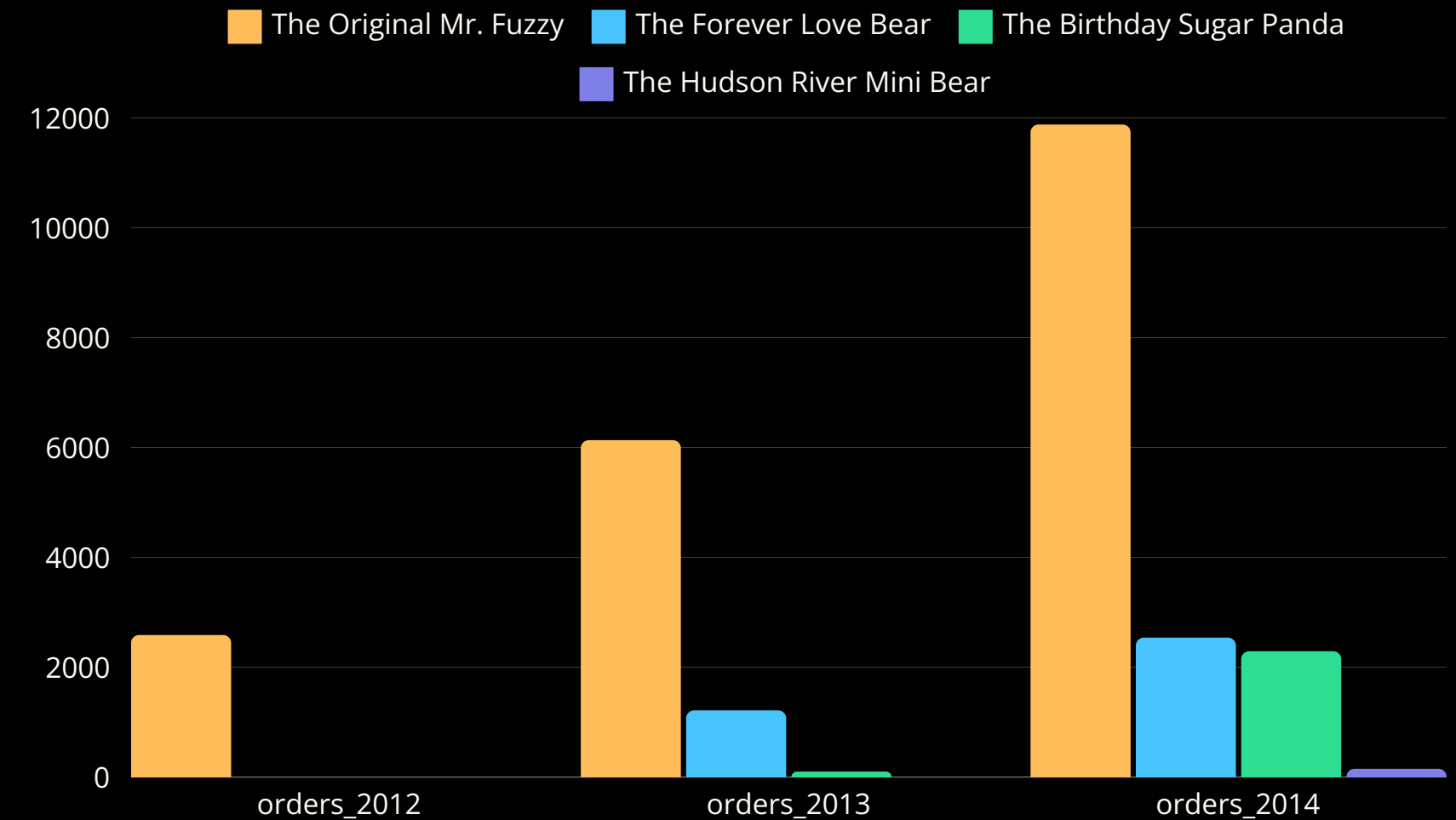
AVG(price_usd - cogs_usd)

Product Business over years

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The Original Mr Fuzzy has maximum overall orders over the years.

- Launched originally in 2012, this product is still top choice of our customers.

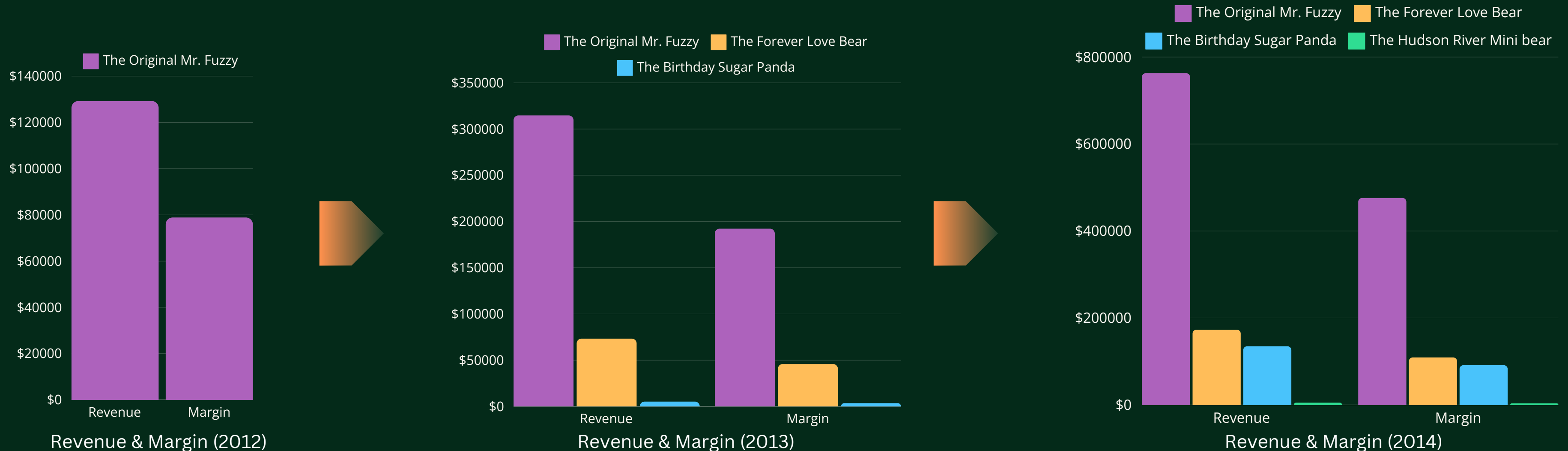


But what about other business metrics?

Revenue, Margin & AOV

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The Original Mr Fuzzy has drives highest revenue and margin overall.

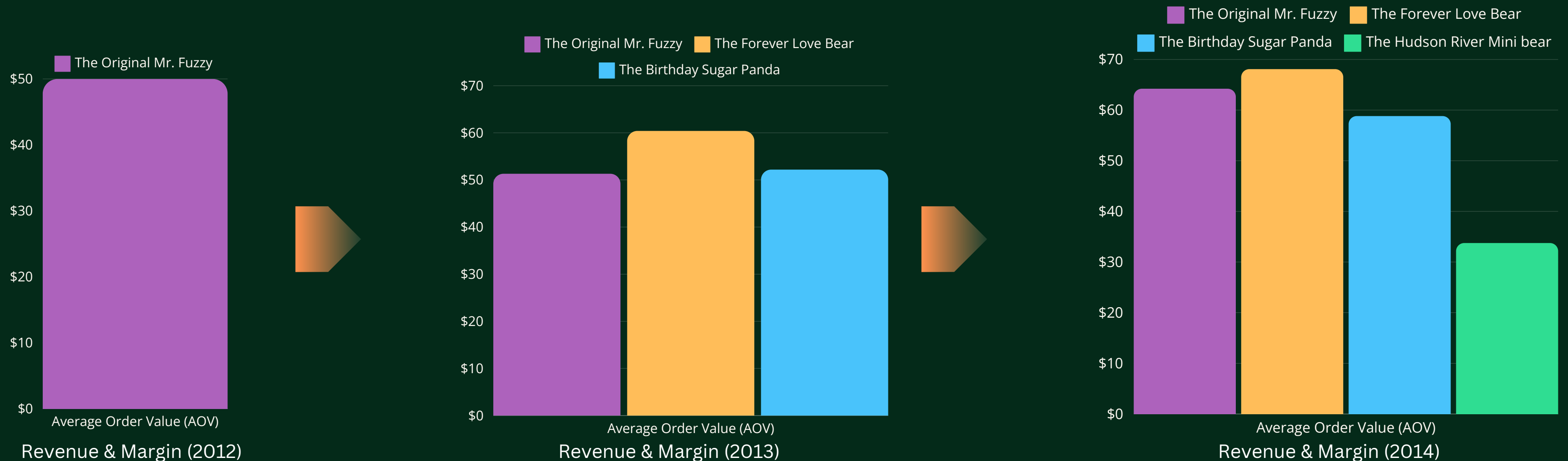


- **“The original Mr Fuzzy”** is the largest contributor of both **Revenue & Margin**.
- **“The Forever Love Bear”** (introduced in 2013) is also a good contributor of revenue and Margin.
- **“The Birthday Sugar Panda”** (introduced in 2013) became a good contributor in 2014 only.

Average Order Value (AOV)

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The Forever Love bear has highest AOV since its year of introduction.



- **“The original Mr Fuzzy”** had AOV of **\$50** during its first year of launch (2012) but did not increase much in 2013.
- **“The Forever Love Bear”** (introduced in 2013) has higher AOV than “The Original...”, i.e., **\$60**.
- **“The Birthday Sugar Panda”** (introduced in 2013) has increased its AOV from **\$52** to **\$59** since its year of introduction.

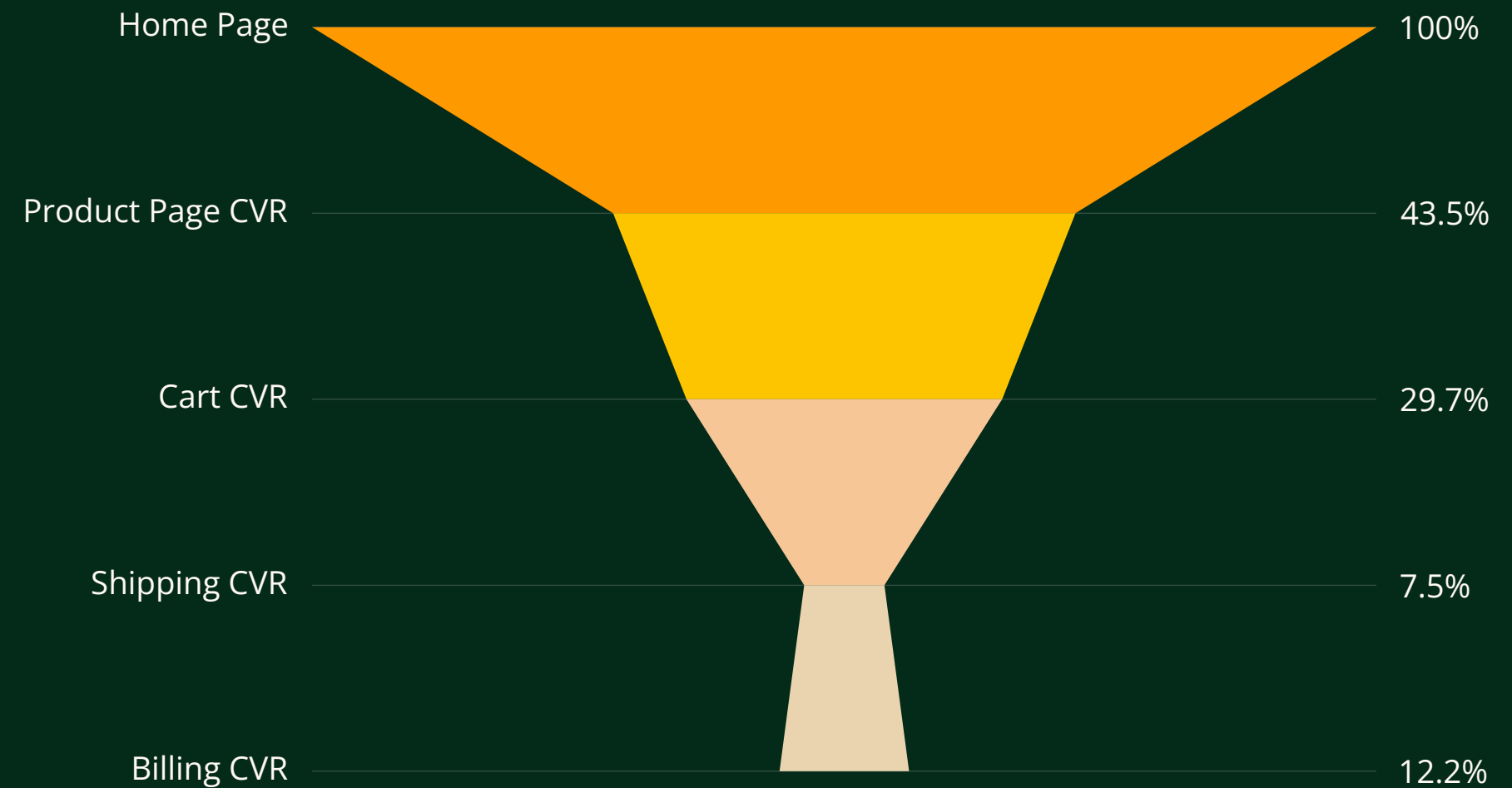
Product Level Conversion Funnel

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We are looking into conversion of sessions as we go till the billing page.

For every **1000 sessions** on **home page**, **43.5 %** go to **The Original Mr Fuzzy**.

- The conversion funnel on right show step by step conversion rates of the sessions.

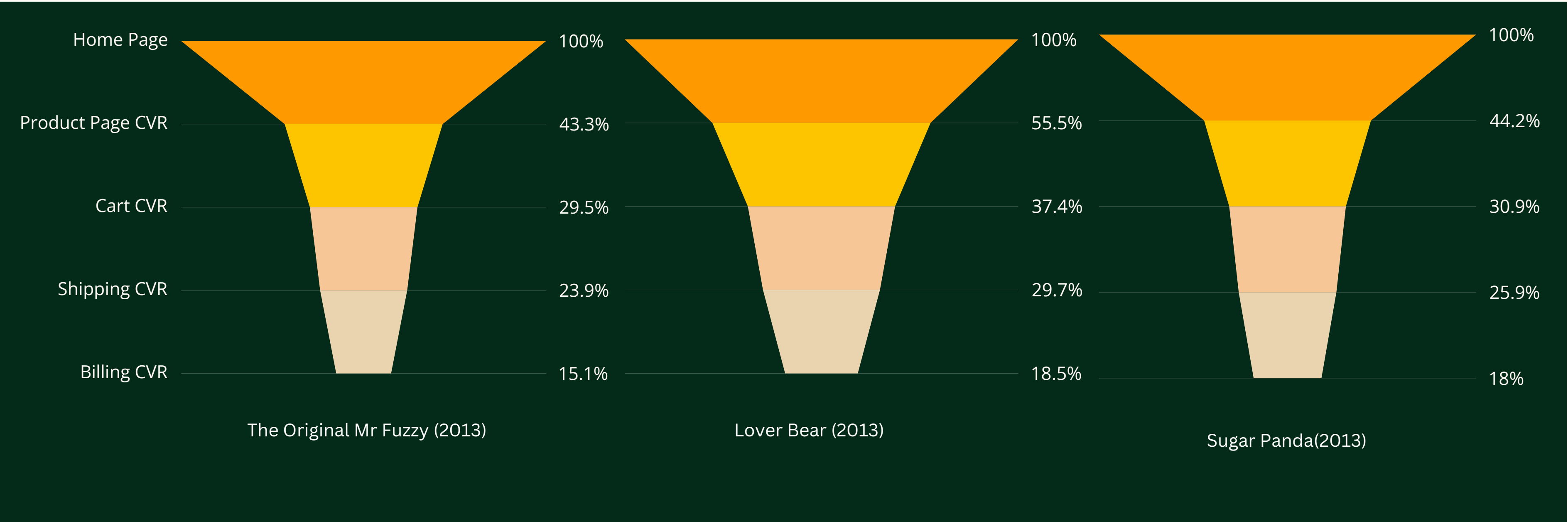


The Original Mr Fuzzy (2012)

Product Level Conversion Funnel

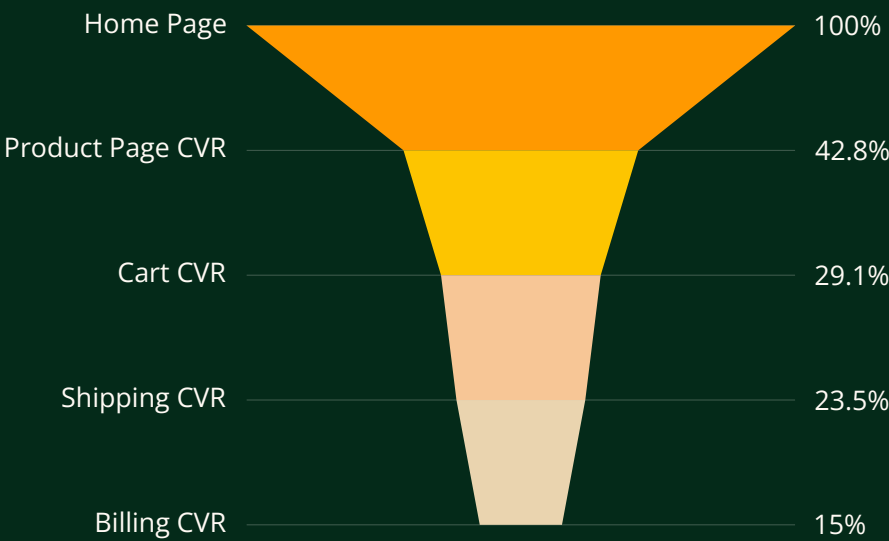
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Love Bear has **highest overall conversion rate** in 2013.

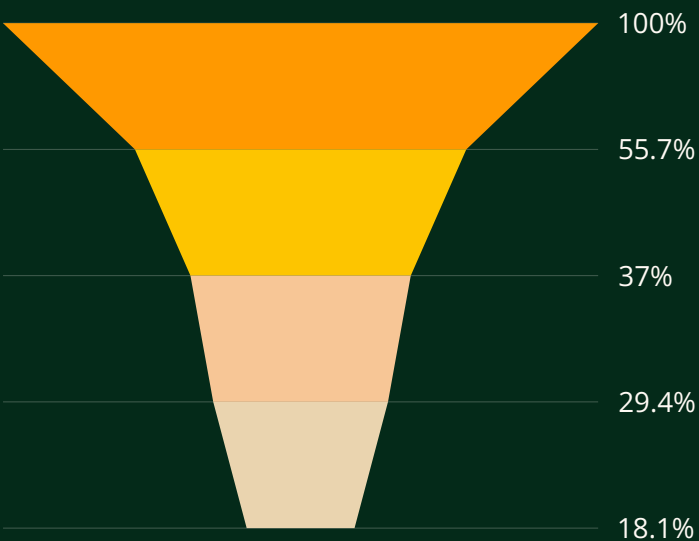


Product Level Conversion Funnel

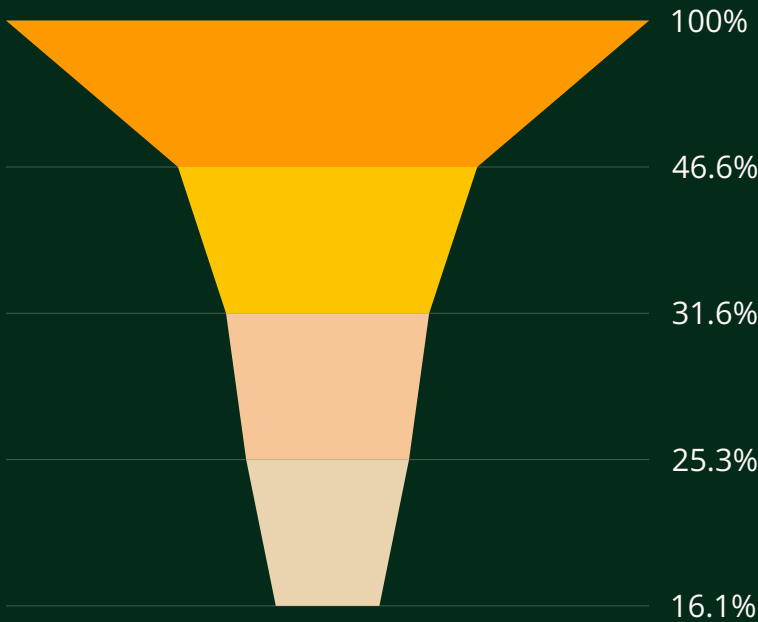
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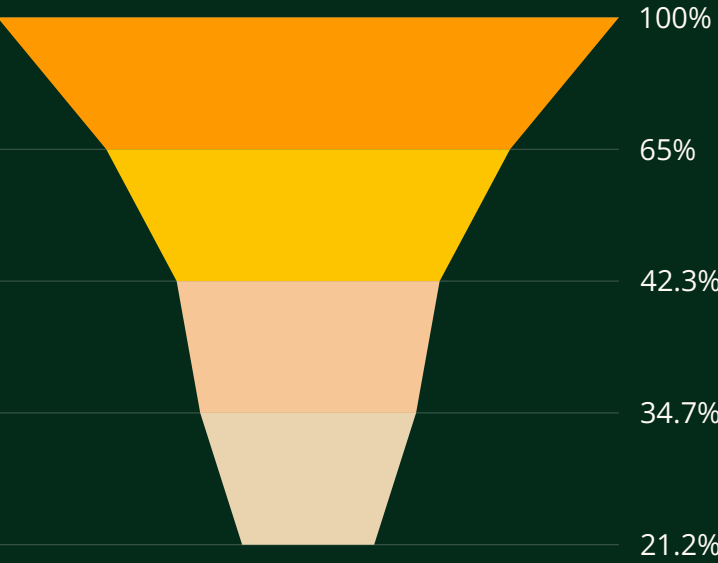
The Original Mr Fuzzy (2014)



Lover Bear (2014)



Sugar Panda(2014)



Hudson Bear (2014)

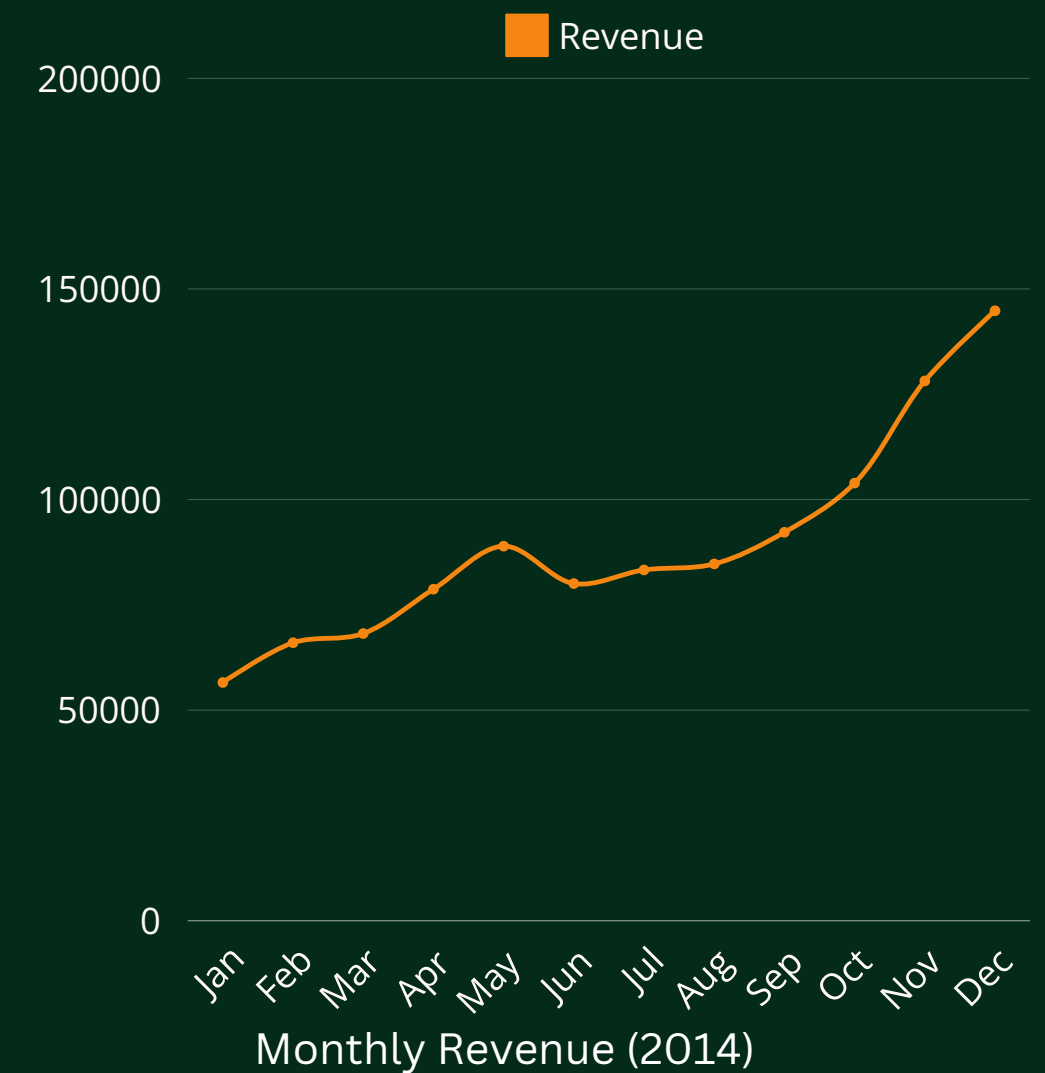
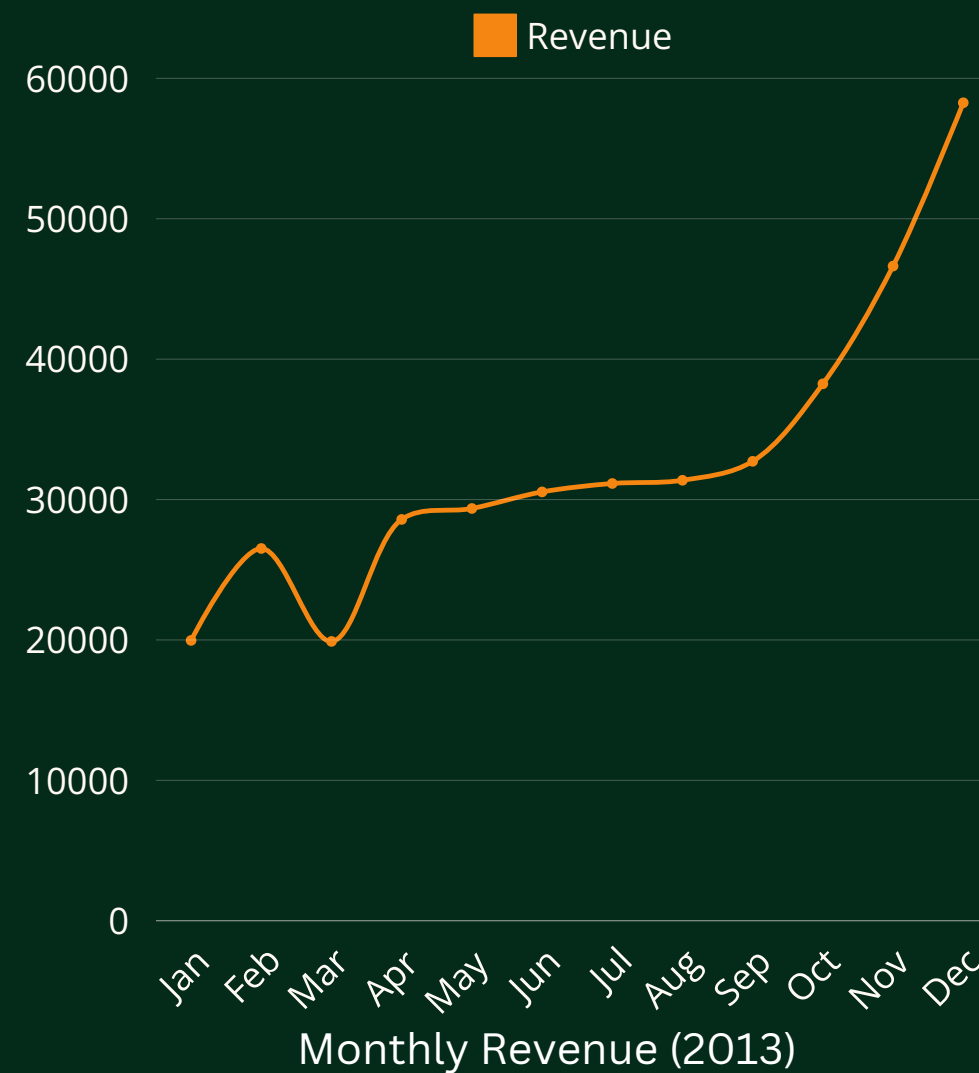
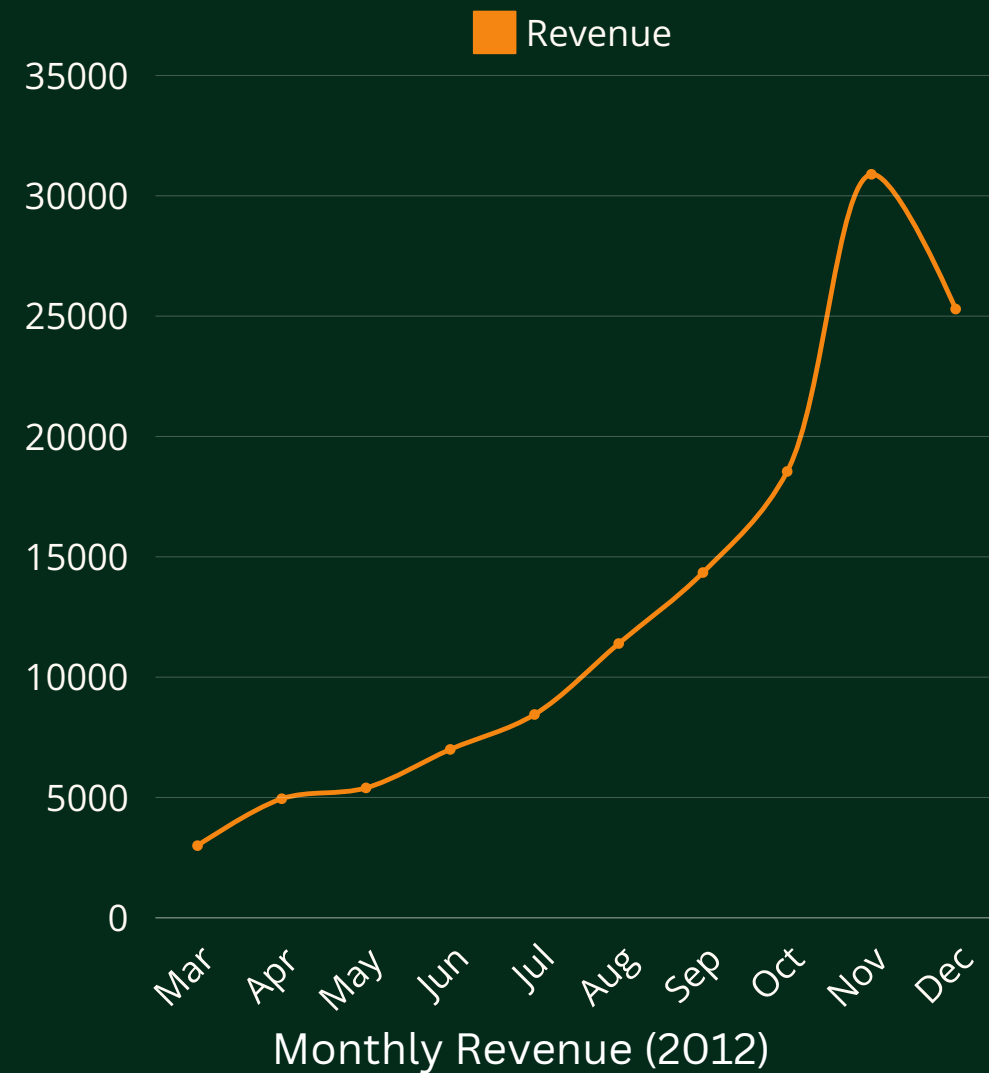
Conversion Funnels are **roughly the same** in next year, i.e., **2014**.

Monthly Sales and Revenue

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We have already seen **sales increasing for each month during website analysis**, let us look at **Revenue and Margin**.

Maximum Revenue in 2014 is 4 times of max revenue in 2012.



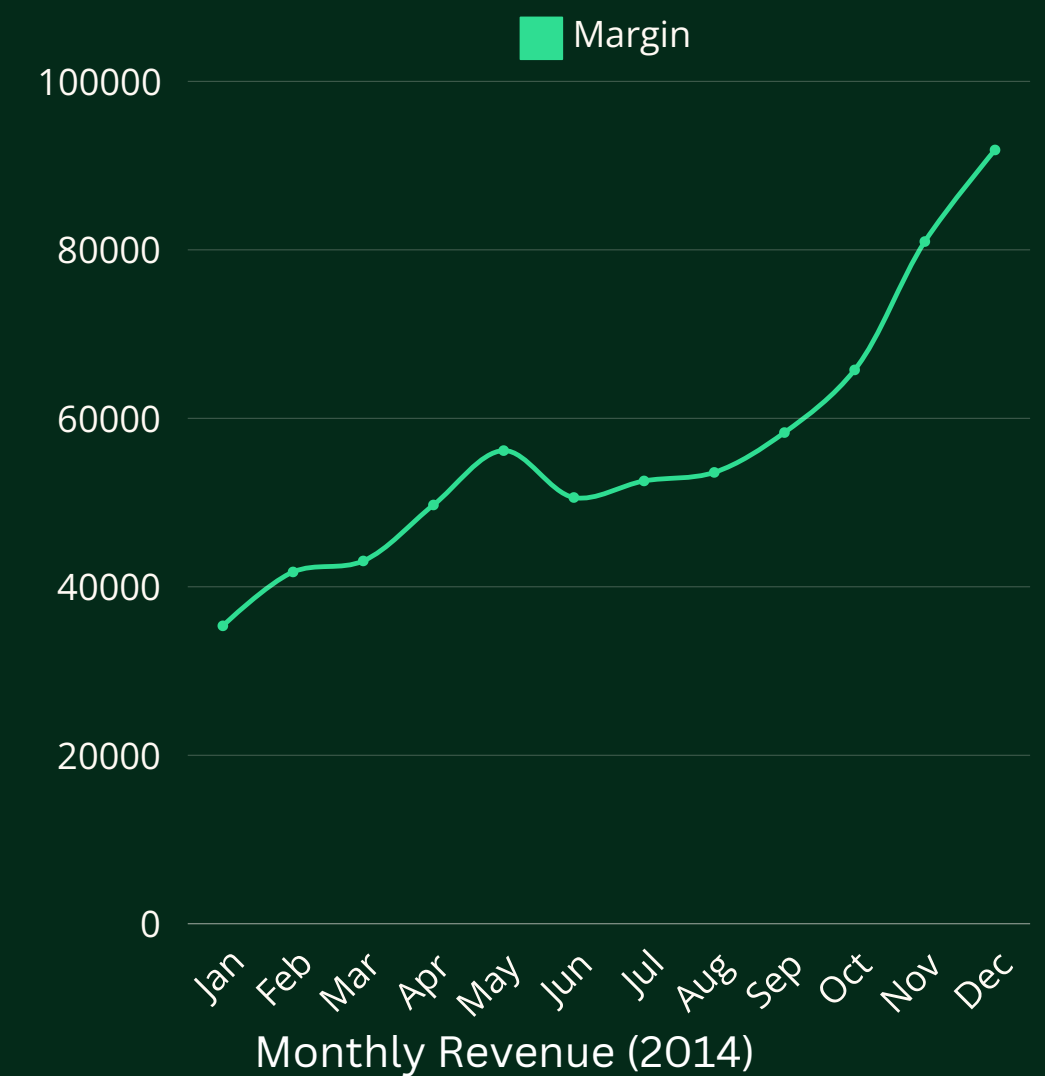
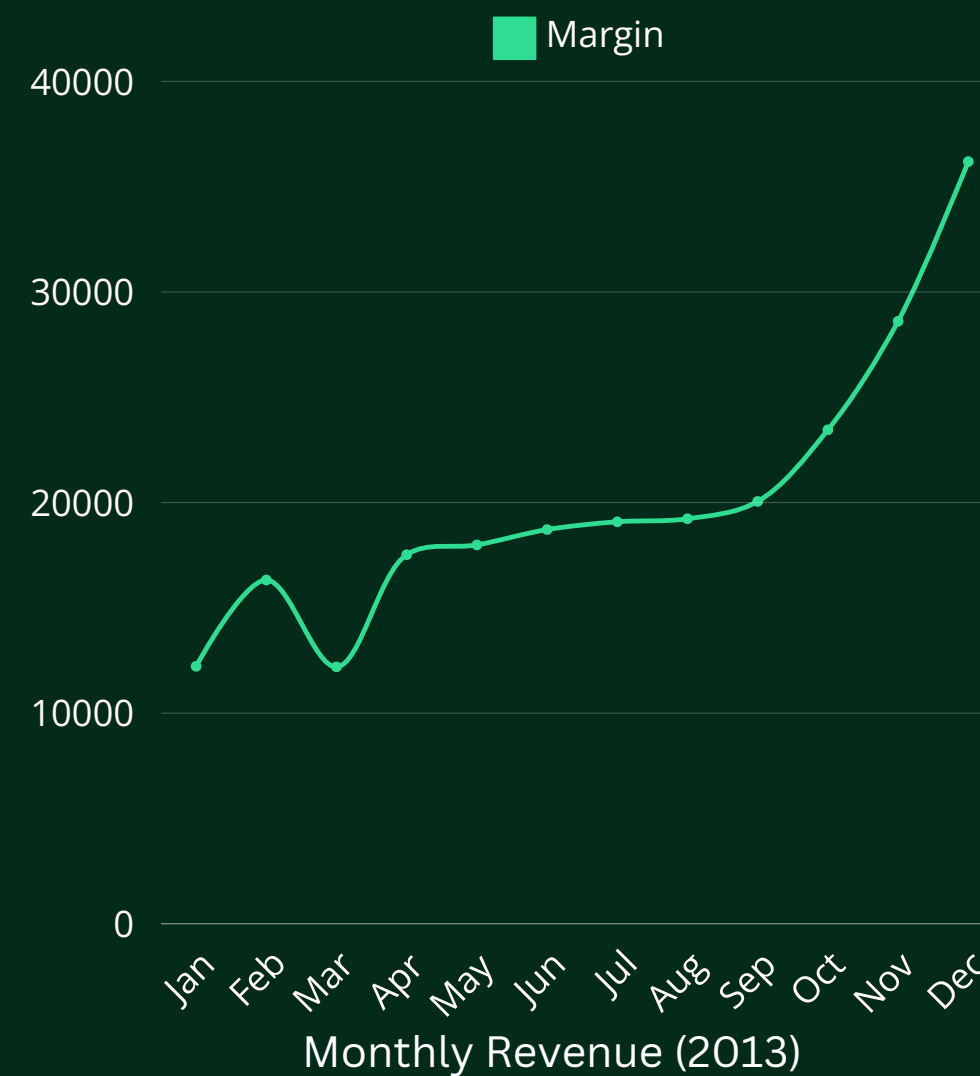
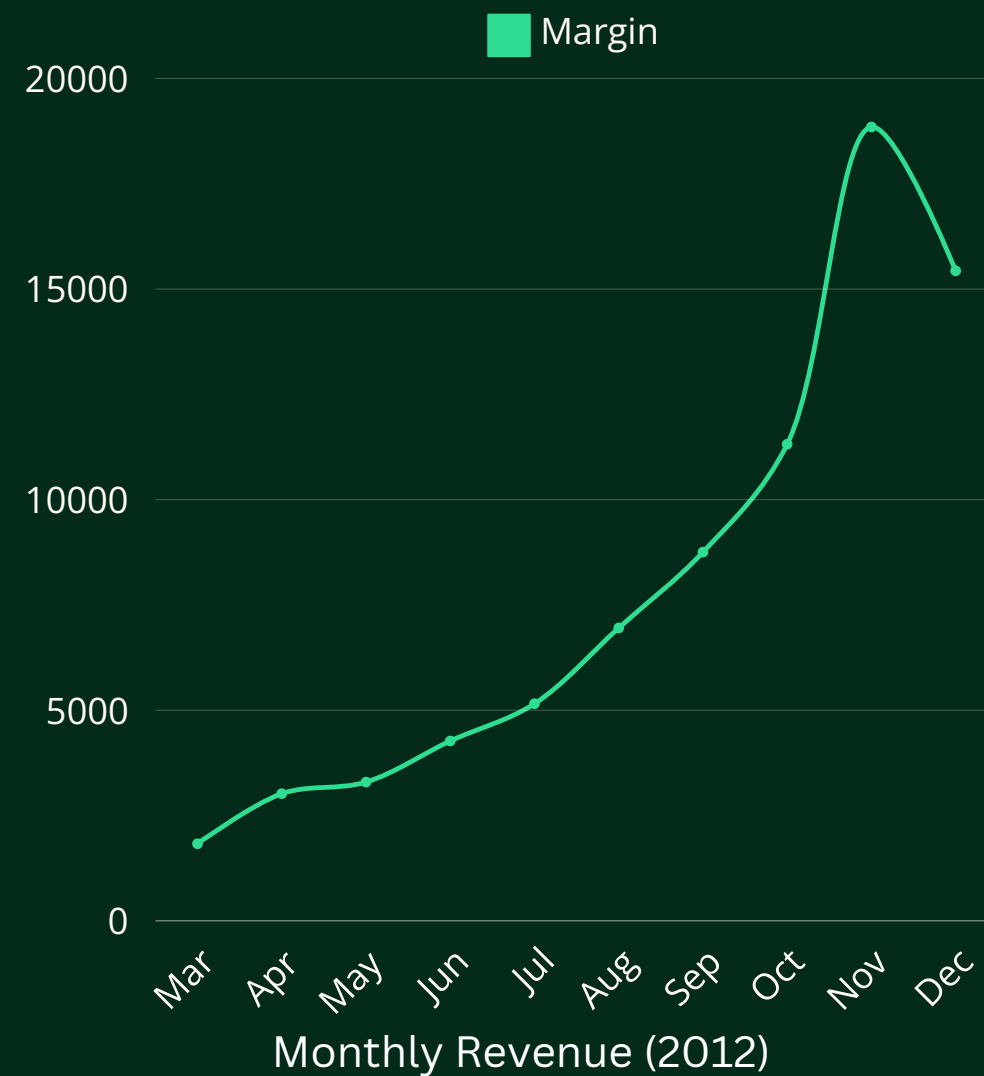
- For 2012, revenue increased over the year becoming maximum revenue (\$31 K) in November (also max orders in this month)
- 2013, similar pattern was observed where revenue maxed out in December (\$58 K)
- 2014, revenue kept increasing over year, peaking in December at \$145 K (more than 4 times of max revenue in 2012)

What about Margin?

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Sales and Revenue are in **favor of the business**, but is it **generating good margin** (profit)?

Maximum Margin in 2014 is 4 times of max margin in 2012.



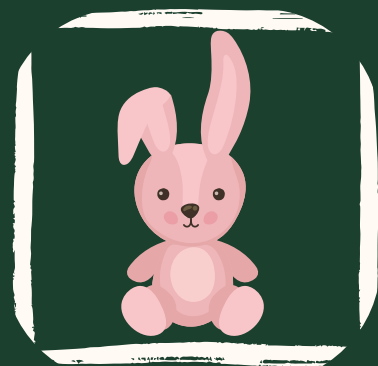
- For 2012, margin increased over the year becoming maximum margin (\$19 K) in November (also max orders in this month)
- 2013, similar pattern was observed where margin maxed out in December (\$36 K)
- 2014, margin kept increasing over year, peaking in December at \$92 K (more than 4 times of max margin in 2012)

Product Level Website Analysis

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Product-focused website analysis is about **learning how customers interact with each of the products** and **how well each product converts customers**.



48% CVR



18% CVR



10% CVR



32% CVR

USE CASES:

- Understanding which products generate most interest on multi product showcase page.
- Analyzing impact on website CVR when new product is added.
- Building product specific conversion funnels to understand whether certain products convert better than others.

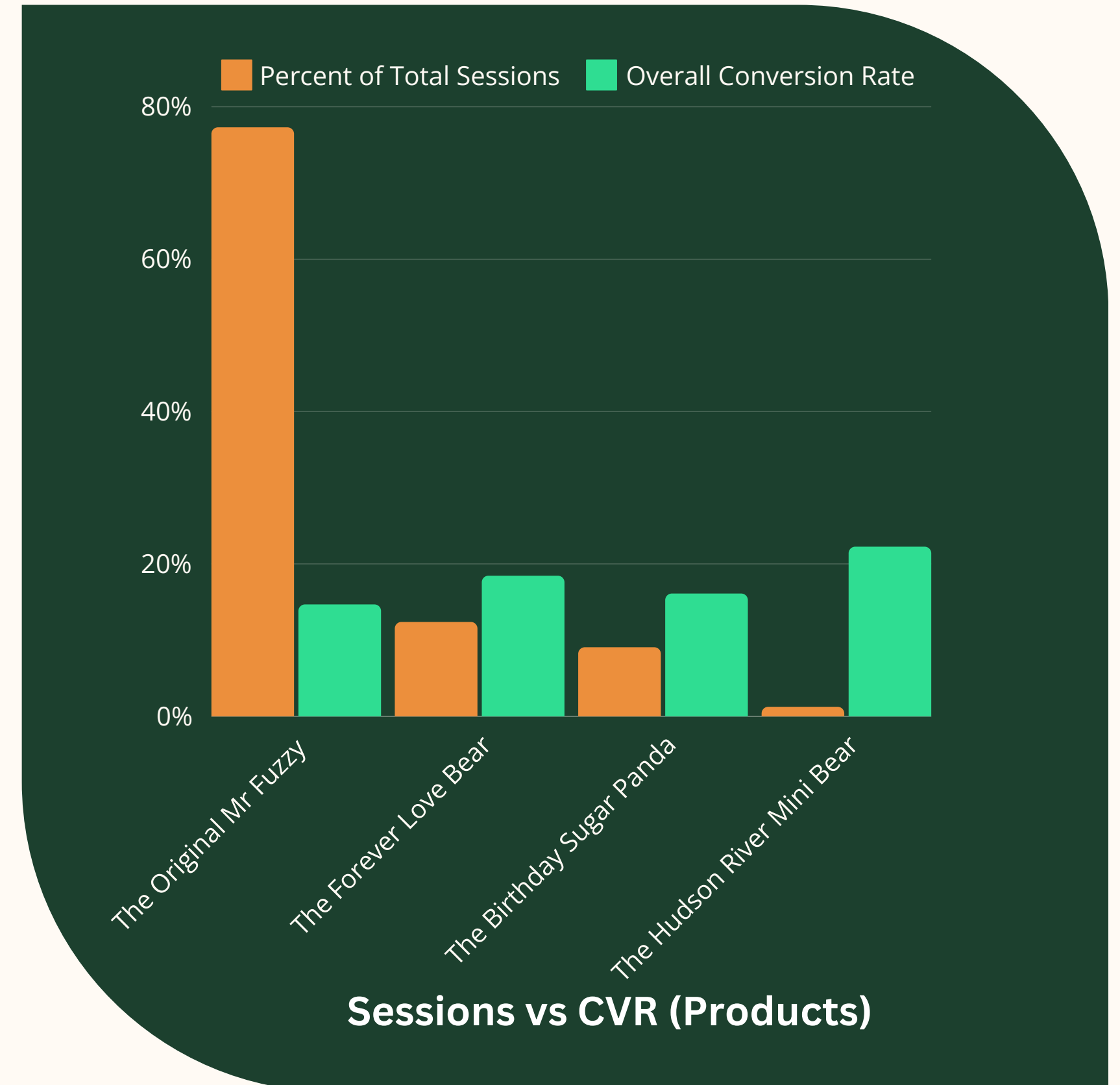
Product, Sessions and CVR

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Let us look into overall sessions and Conversion Rate (CVR) for each product in our inventory.

The Original Mr Fuzzy has **highest total sessions** but **The Hudson River Mini Bear** has **highest CVR**.

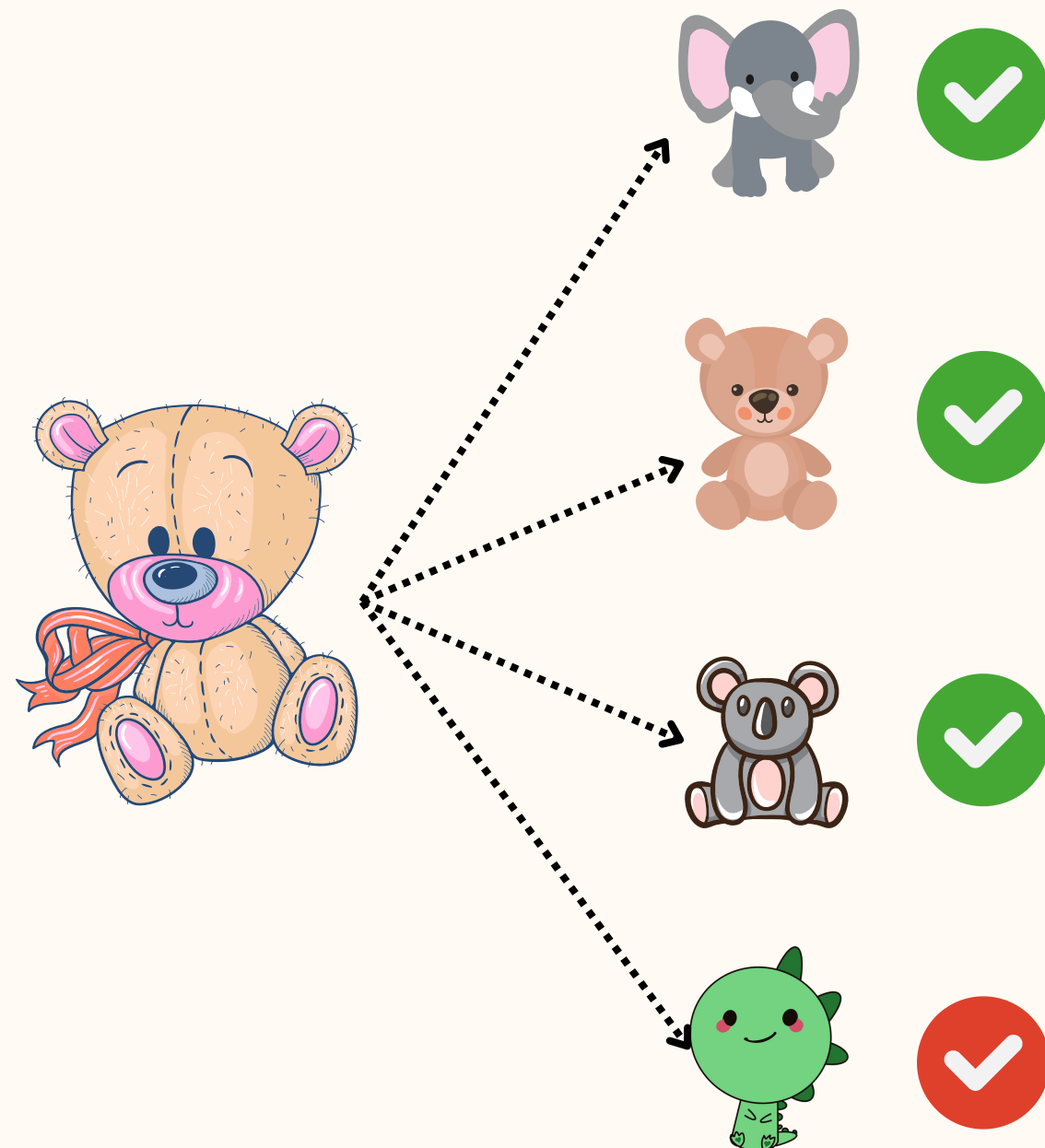
- **77%** of total sessions go to **The Original ...** and has a good **CVR of 15%**
- **The Hudson....** has **lowest sessions** but for some reason has **highest CVR**.
- So, it would be a **good strategy to look deeper into why CVR is highest** here and apply it on other products.



Cross-Selling Analysis

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💰 Cross sell Analysis is about **understanding which products users are most likely to purchase together** and **offering smart product recommendations**.



USE CASES:

- Understanding which products are often purchased together
- Testing and optimising the way cross sell products on your website.
- Understanding the conversion rate impact and the overall revenue impact of trying to cross sell products.

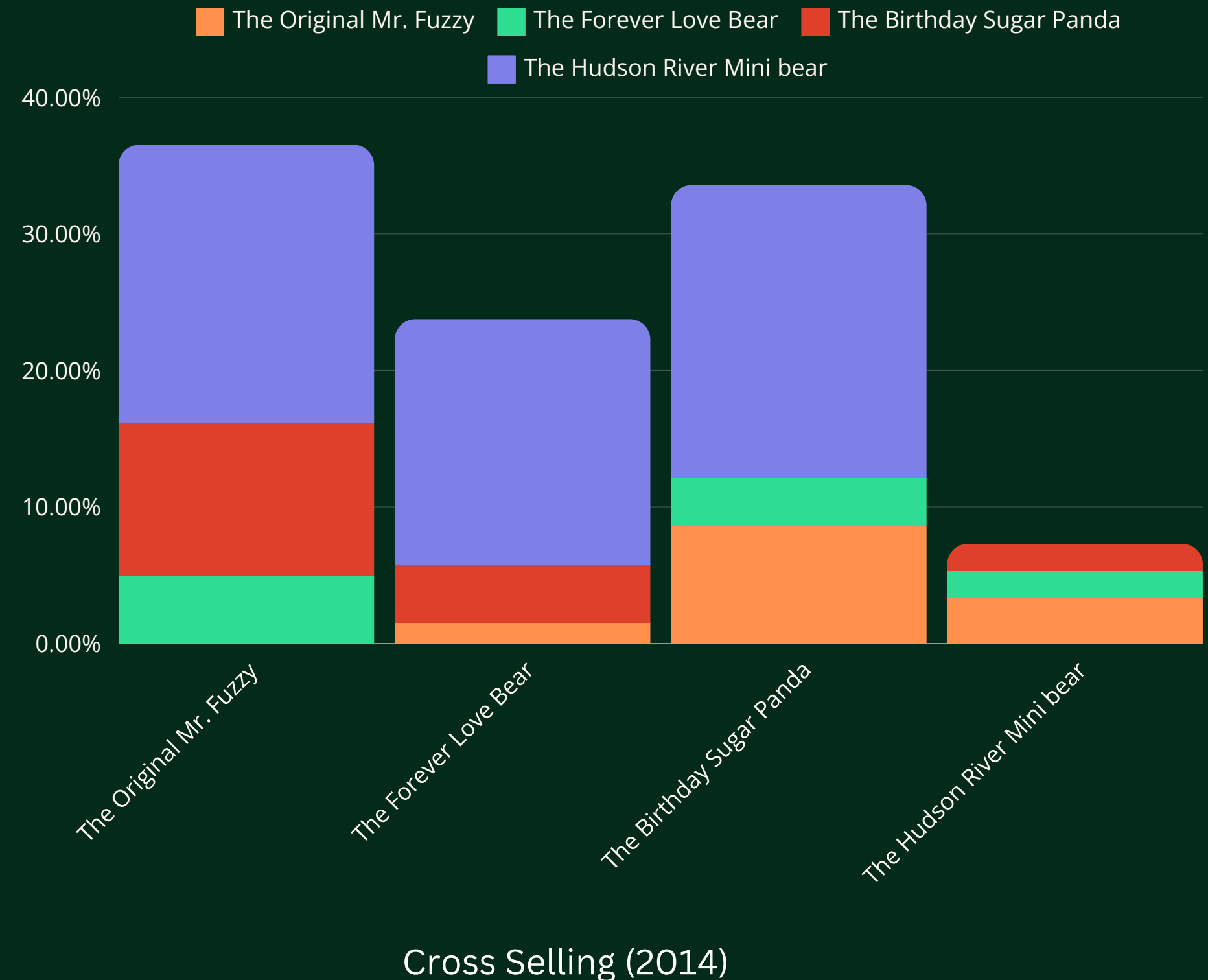
Cross Selling across products

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- Since, all four products got fully available for sale in 2014, we are going to look at 2014 and 2015 year.

The Hudson River Mini Bear is the **most cross sold item** across all products in 2014

- When either of the three products ("The Original...", "The Forever...", "The Birthday...") is sold, there is a 20% (approx.) chance of "The Hudson..." getting sold too.
- There is no significant change in cross selling rates for 2015, so we are not looking at that.



Product Refund Analysis

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Analyzing product refund rates is about **controlling for quality** and **understanding where we might have problems to address**



USE CASES:

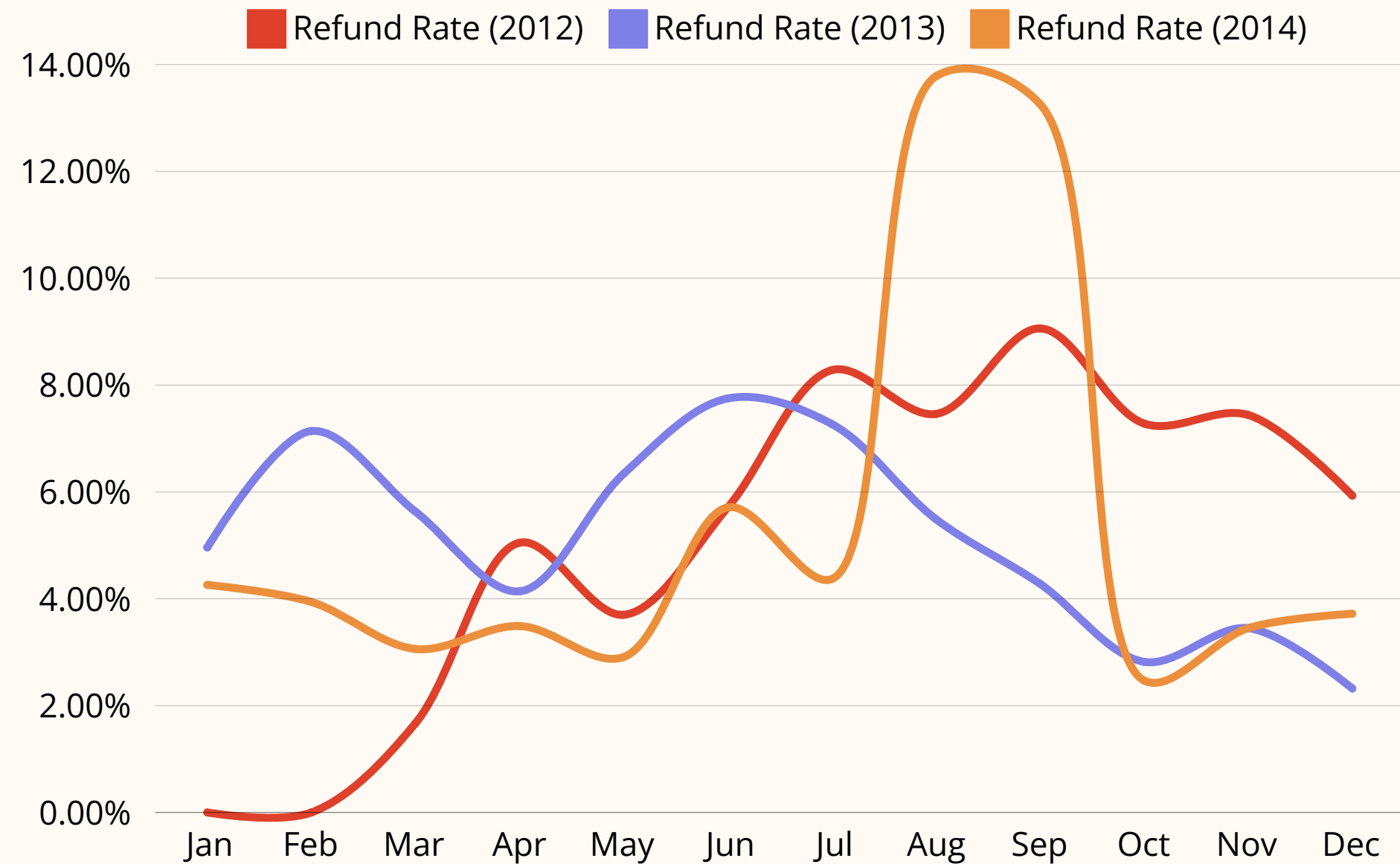
- Monitoring products from different suppliers
- Understanding refund rates for products at different price points
- Taking product refund rates and associated costs into account when assessing overall performance of business.

Refund Analysis (The Original...)

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The Original Mr Fuzzy has **higher refund rates** towards the **end of the year**

- “The Original...” has refund rates in **2%-8%** range (except in 2014).
- Look deeper into 2014 (Aug and Sep) and check what went wrong.



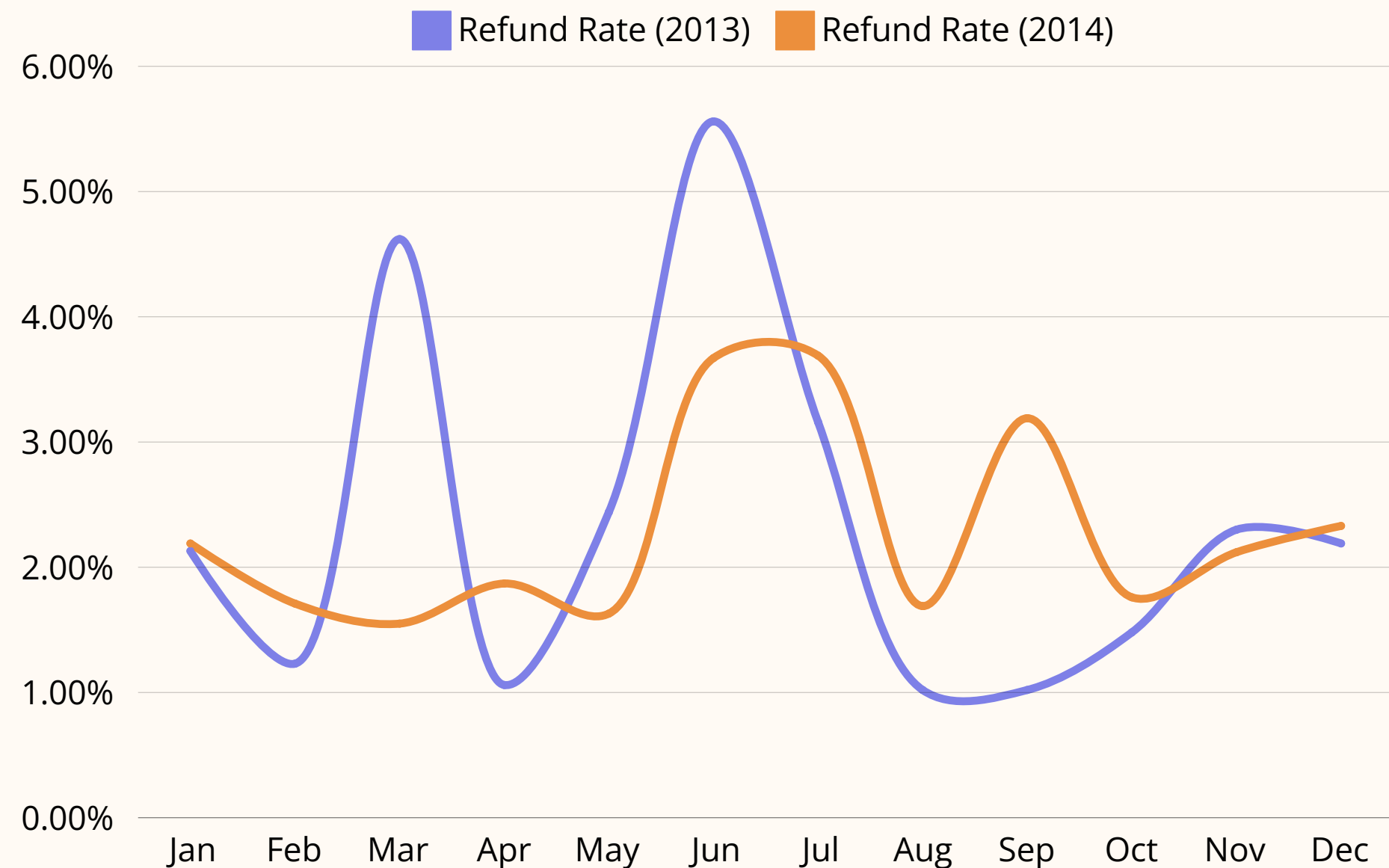
The Original Mr Fuzzy Refund Rates

Refund Analysis (The Forever...)

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The Forever Love Bear has **higher refund rates** which **peak twice a year**

- “The Forever..” has refund rates in **2%-5.5%** range (**2013**) and **2%-3.7%** (**2014**).
- We observed **two peaks: Mar and Jun (2013)** and **Jun and Sep (2014)**



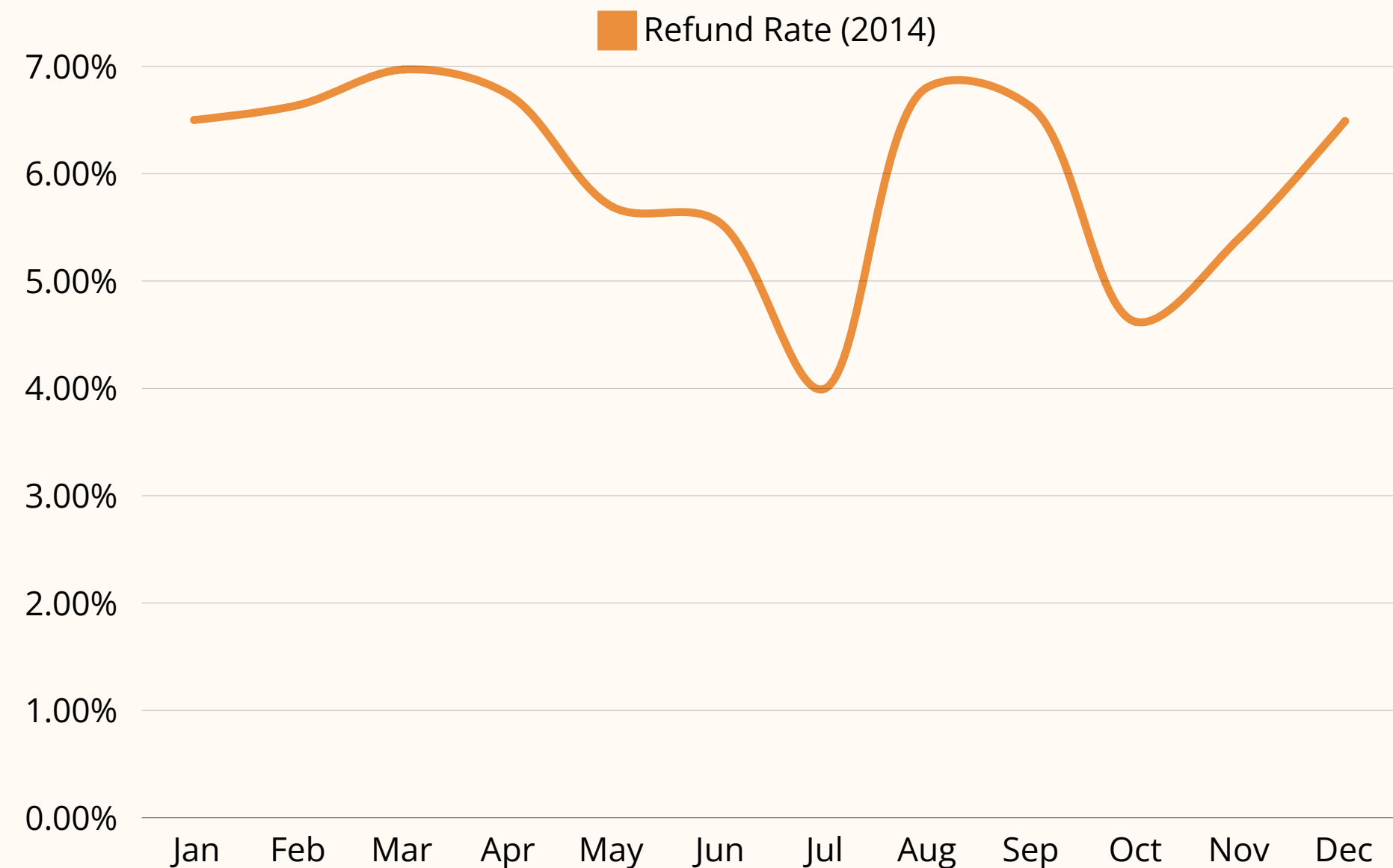
The Forever Love Bear Refund Rates

Refund Analysis (The Birthday..)

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The Birthday Sugar Panda has **high refund rates** which **peak during several months**.

- “The Birthday..” has refund rates in **4%-7%** range (**2014**).
- Refund rates remain **roughly same** within **5% to 7%** range only **minimising to 4% in July**.



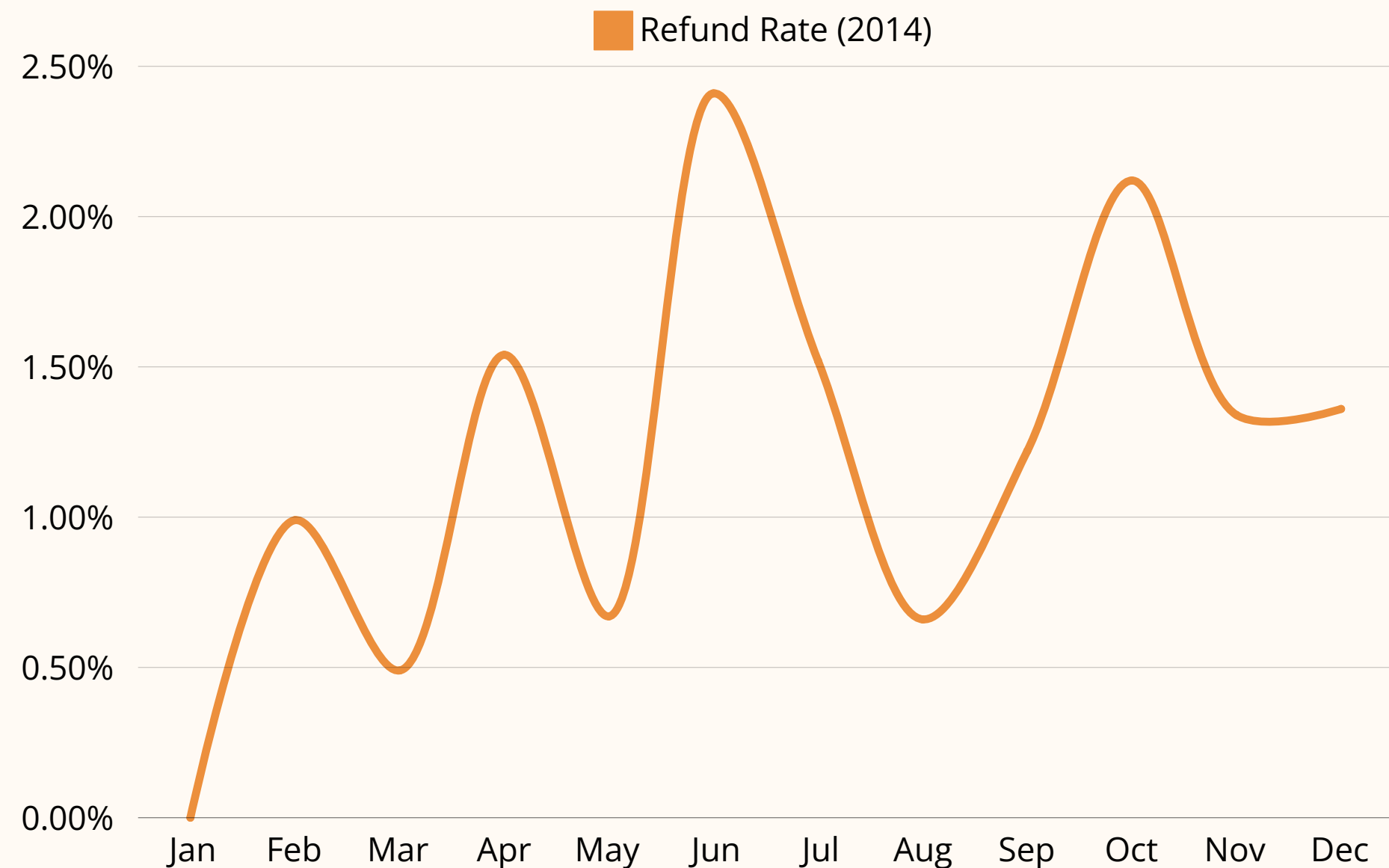
The Birthday Sugar Panda Refund Rates

Refund Analysis (The Hudson..)

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The Hudson River Mini Bear has so far the lowest refund rates.

- “The Hudson..” has refund rates in **1%-2.5%** range (**2014**).
- **Refund rates** keep fluctuating but are **not drastically high**.
- But this **might be due to number of orders not being very high**. Monitor it closely as the number of orders increases over years.



The Hudson River Mini Bear Refund Rates

05

User Analysis

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- [Introduction](#)
- [Repeat Sessions over years](#)
- [Entry Channels for New Sessions](#)
- [Entry Channels for Repeat Sessions](#)
- [Conversion Rates and Revenue Per Session](#)

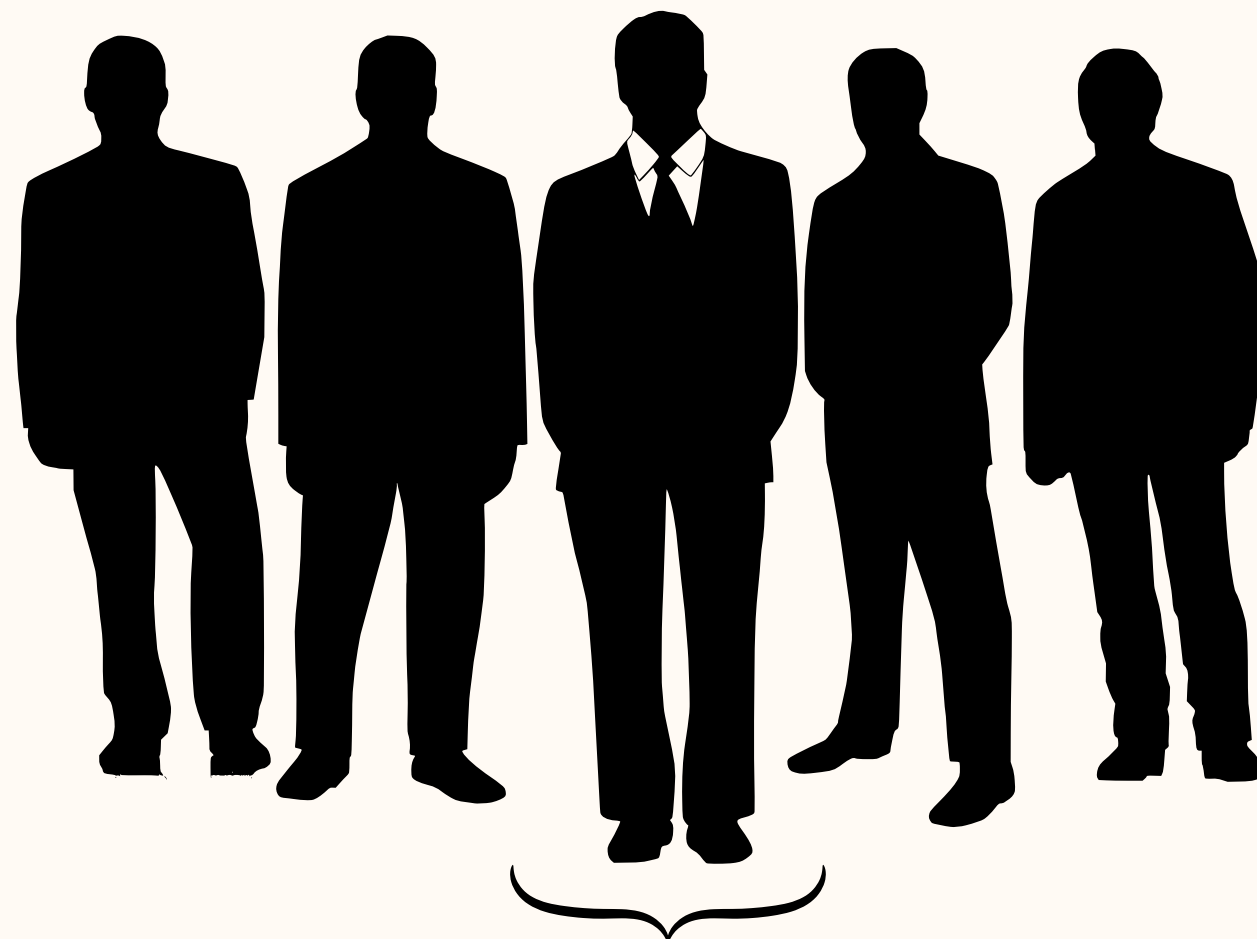


Analyze Repeat Behaviour

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Analyzing repeat visits helps in **understanding user behaviour** and **identify some of our most valuable customers**.



30 Sessions
10 Transactions
\$15 AOV
42% CVR

USE CASES:

- Analyzing repeat activity to see how often customers are coming back to visit website.
- Understanding which channels they use when they come back and whether or not we are paying for them again through our paid channels.
- Using repeat visit activity to build a better understanding of value of a customer in order to better optimise marketing channels.

Tracking Repeat Customers

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🔄 To track this repeat customer behaviour, businesses use **browser cookies**.

How?

Cookies have **unique ID values** associated with them, which allows businesses to **recognise a customer when they come back** and **track their behaviour over time**.



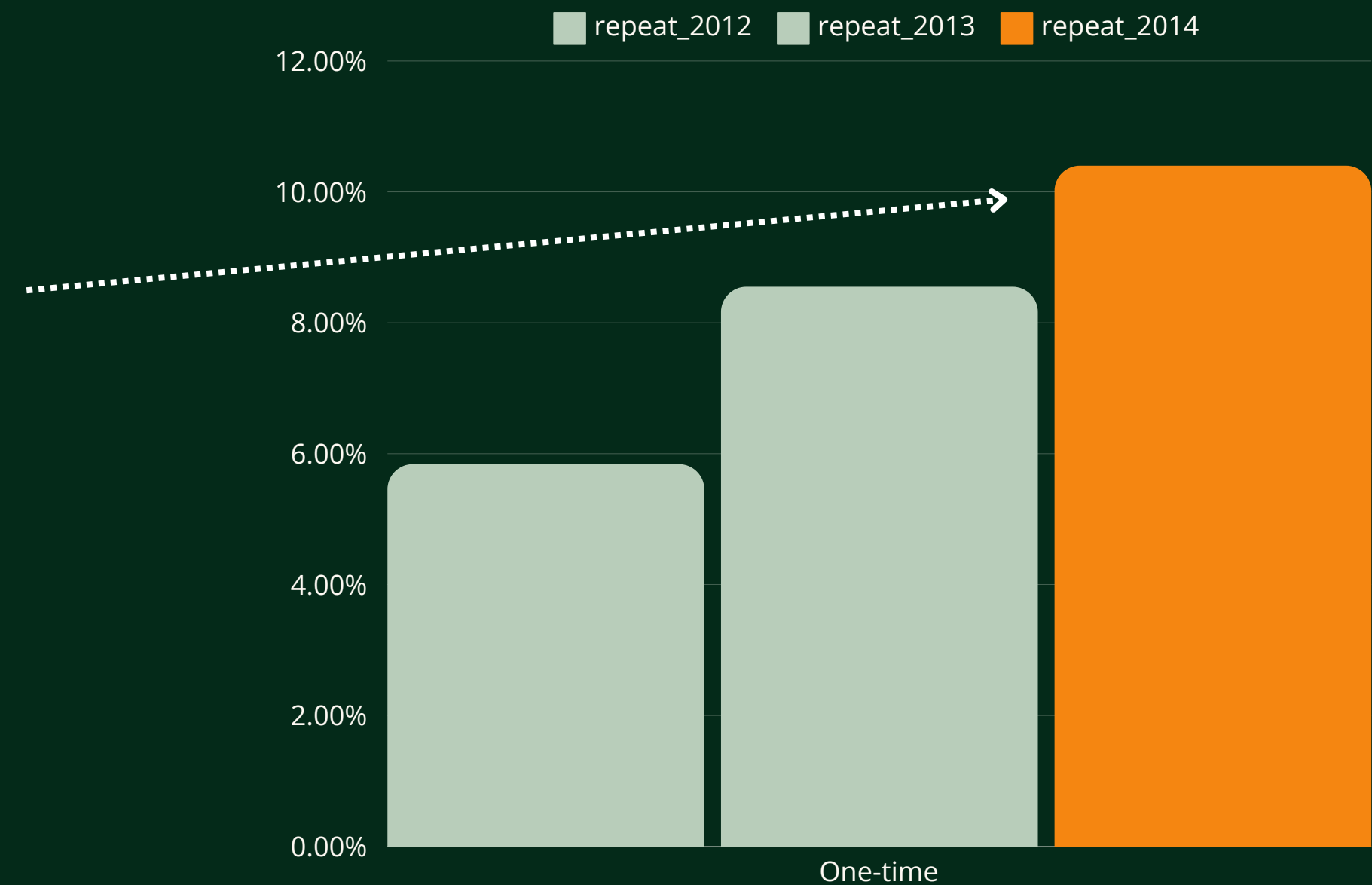
Repeat Sessions over years

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Number of **Repeat Sessions** have **doubled** since the business started in 2012.

In **2012**, only **5.84%** of sessions repeated but by **2014**, **10.40%** of total sessions are repeated.

On an average, for all repeat customers, it takes them **33 days** to make another visit to our website.



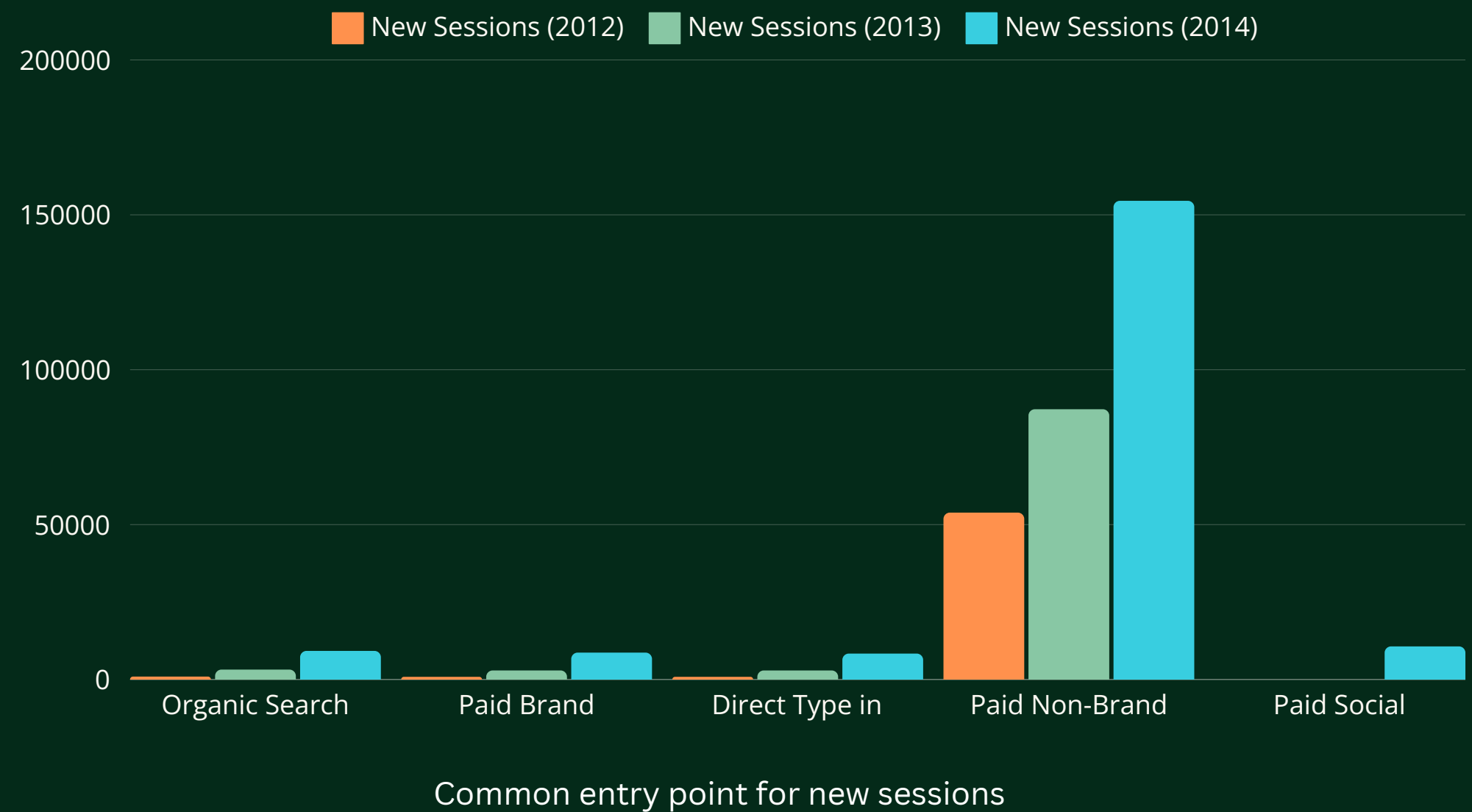
Entry Channels for New Sessions

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- There are customers who come to our website again in an average 33 days, but what channels do they use to arrive our website?
- But let us first look at the common entry points for our new customers.

Paid Non-brand is the most popular channel for new website sessions.

- Paid nonbrand channel brings most of the new sessions and over the years, number of sessions keeps increasing.



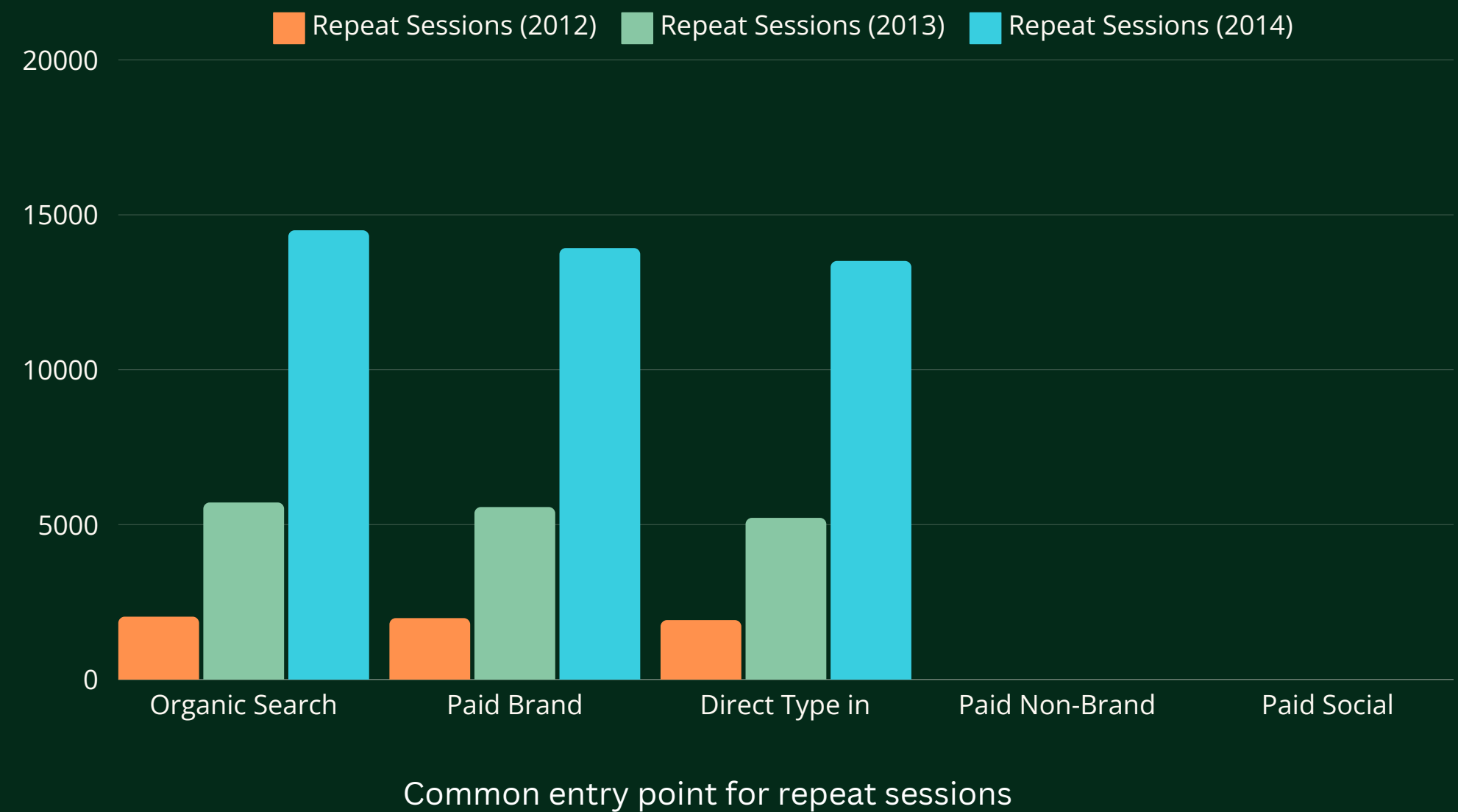
Entry Channels for Repeat Sessions

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Paid nonbrand has been great in driving new sessions to the website. But what about repeat sessions?

Paid Non-brand brings no repeat sessions to the website.

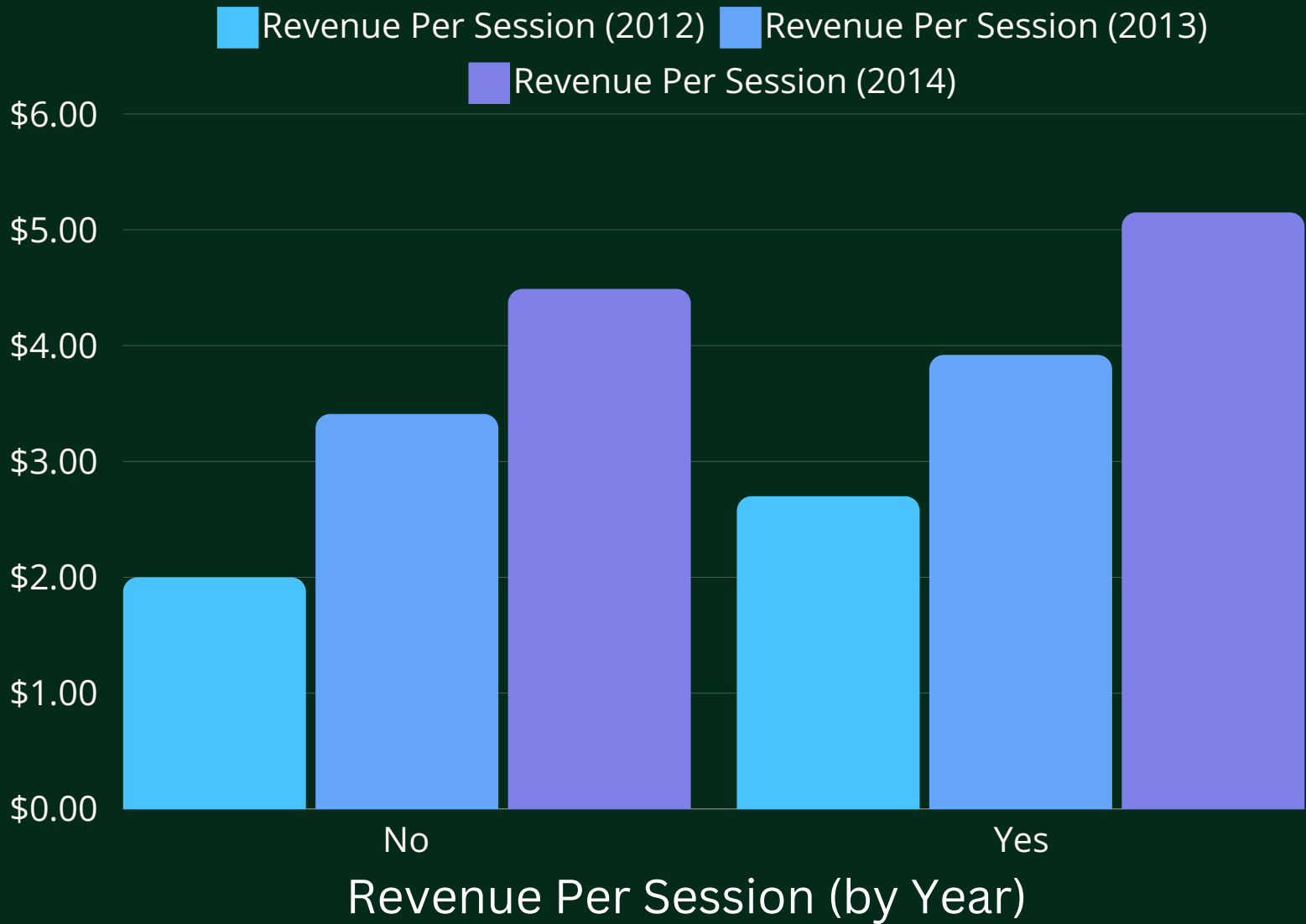
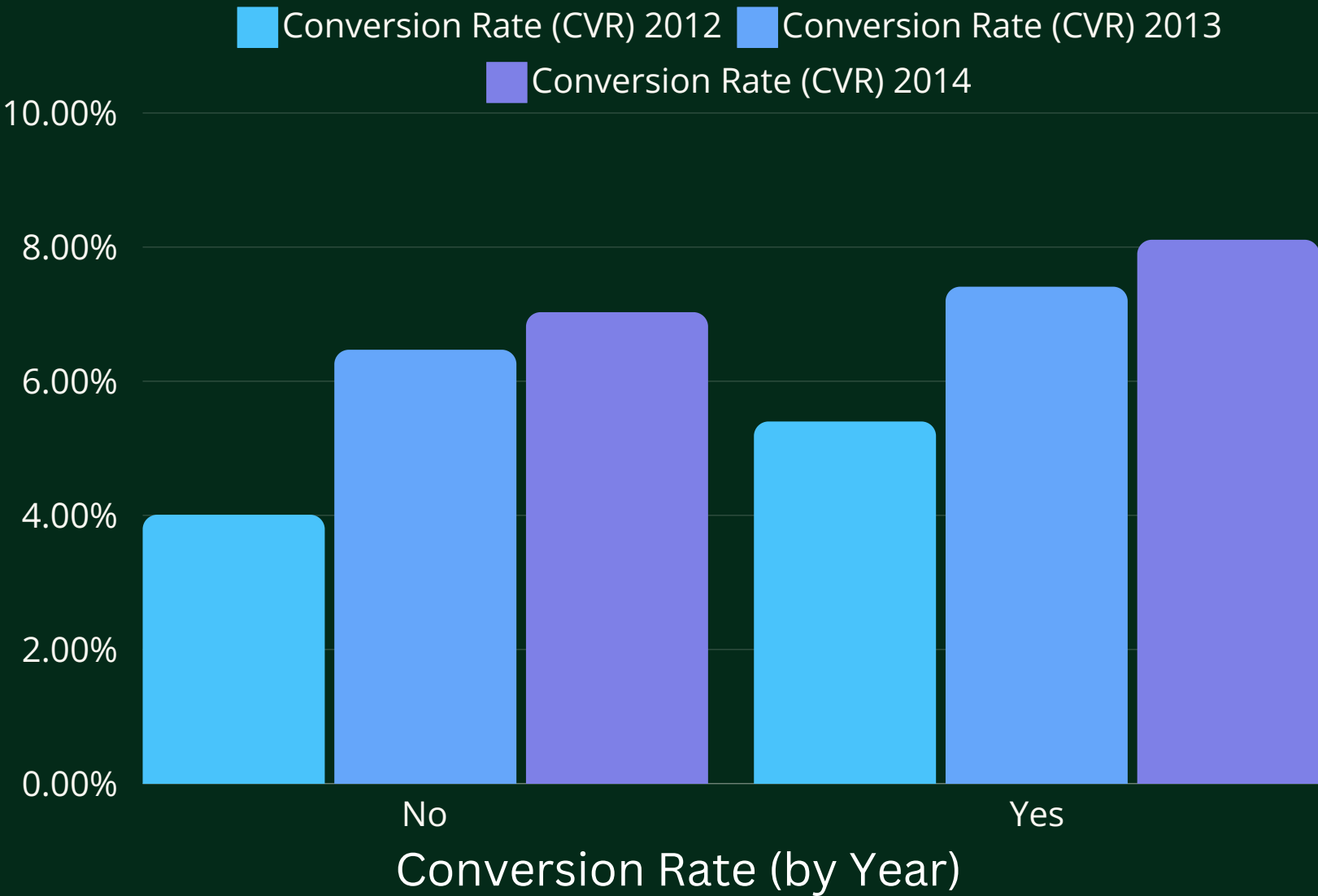
- All repeat traffic is coming through **organic** (non paid traffic), **direct type in** (directly to website) and **paid brand** (which is much cheaper than non brand).
- This means, **we are not spending too much money** as paid nonbrand is supposed to bring new sessions (not repeated ones).



Conversion Rates and Revenue per Session

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Both **CVR** and **Revenue Per Session** are **higher for repeat session**.



- New Sessions (CVR) -----> 4% (2012) to 7% (2014) (+57%)
- Repeat Sessions (CVR) -----> 5.40% (2012) to 8.11% (2014) (+67%)
- New Sessions (Revenue Per Session) -----> \$2 (2012) to \$4.49 (2014) (+45%)
- Repeat Sessions (Revenue Per Session) -----> \$2.70 (2012) to \$5.15 (2014) (+52%)

Conclusion

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01

Traffic Source Analysis

- Understood where our customers are coming from.

02

Website Performance Analysis

- Understood overall performance of our company website.

03

Analyzing Business Patterns & Seasonality

- Understood business throughout the year, key months and ones to focus more on.

Conclusion

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04

Product Analysis

- Understood products sold and their contributions to overall business.

05

User Analysis

- Understood new and repeat customers closely.

Thank
you

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